MOVING AND FILTERED LESLIE MODELS: RIGID CRITICAL GEOMETRY, NEGATIVE ARITHMETIC-FREE AUDIT, DIVISIBILITY FILTRATIONS, AND COHERENT ARITHMETIC-DRIVEN TOWERS

Abstract. This version separates three questions that must not be conflated: static spectral completion, exact recurrence compatibility with the observed scalar sequence, and dynamical transport between overlapping windows. For a fixed Perron value b, K prescribed critical conjugate pairs, and a fixed auxiliary degree q, the repaired coefficient family has ambient degree $D = 2K + 1 + q$ and, at regular points, dimension $K + q$ and codimension $K + 1$, independently of q. For order-D recurrence windows displaced by s samples, the exact common-equation slice has generic local dimension $\max\{0, s - K - 1\}$. Hence the first generic one-dimensional repaired motion still occurs at $s_* = K + 2$. The repaired update is generally a nonlinear curve rather than the rigid affine line. If the first-new-value observation is transverse to this curve, that single scalar selects the local next coefficient vector and the remaining $K + 1$ new values are then fixed recursively.

The auxiliary multiplier coefficients are therefore not independent innovations under the fixed-(K, q, b) regular continuation hypothesis; they become hidden channels only if a completion, branch, q, or b is reselected using new information. Static Toeplitz completion alone does not establish a data-compatible moving realization. A recurrence-first, frequency-free membership test is consequently made primary. The numerical audit performed on 5 September 2026 is retained as a negative baseline for the then-current approximate pipeline, while exact conclusions are restricted to the hypotheses actually verified.

Version 8 retains all previous geometric and numerical results but changes the active research direction after a stronger predictive audit. For the frozen arithmetic-free protocol $K = 4$, $b = 1.02$, and $q = 0, \dots, 6$, exact repaired membership is absent, the corrected positive nonlinear approximation does not outperform a direct autoregressive baseline, and phase-randomized surrogates remove the apparent innovation advantage. This closes that *arithmetic-free* route for the frozen protocol without asserting a universal impossibility for Leslie matrices.

The continuation therefore injects arithmetic before asking Leslie geometry to compress it. The arithmetic source is the finite divisibility poset. Its incidence matrix $D_X = (\mathbf{1}_{d|n})_{n,d\le X}$ is parameter-free and satisfies $D_X \mu_X = e_1$ and $(D_X^{-1})_{n,d} = \mu(n/d)$ when $d \mid n$. A filtration by $\Omega(n)$ yields exact block extensions

$$D_{k+1} = \begin{pmatrix} D_k & 0 \\ B_k & I \end{pmatrix}, \qquad \mu_{k+1} = \begin{pmatrix} \mu_k \\ -B_k \mu_k \end{pmatrix},$$

so every old value is preserved and all new arithmetic information enters through the new incidence block B_k. The active problem is no longer to make a fixed Leslie family discover Möbius without arithmetic, but to determine whether a coherent tower of positive Leslie companions can represent these exact divisor-layer extensions without refitting old levels.

Version 9 sharpens both sides of this bridge. On the Leslie side, the differential of the repaired multiplication map is computed exactly: at fixed b its rank drops precisely when the auxiliary factor contains a critical quadratic factor, not merely when it shares the Perron factor. The regular codimension $K + 1$ is thereby identified as an intrinsic normal-space dimension, observability is reduced to an augmented-rank certificate, and coordinate truncation of a strict Leslie companion tower is shown to add only zero eigenvalues. On the arithmetic side, the new divisor block $B_k(X)$ has rank growing without bound with X, and the raw Möbius layer does not lie in any fixed finite-dimensional linear subspace. Consequently, any genuine low-dimensional Leslie compression must occur only after a causal observable $\mathcal{O}_{k,X}$ acts on the arithmetic layer. The central open condition is therefore not that B_k itself be low rank, but that its observable image satisfy the $K + 1$ normal compatibility conditions and, for exact scalar compression, have rank one along the continued Leslie branch.

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Remark 0.1. This document is a research notebook. The algebraic identities, the rigid overlap theorem, the repaired local-dimension theorem, and the local one-scalar continuation theorem are proved only under their stated rank, regularity, fixed-degree, fixed-Perron, and branch hypotheses. The proposed connection with consecutive windows of the Mertens function remains an experimental programme, not a proved arithmetic theorem. In particular, static spectral completion does not imply that the completed polynomial is the recurrence polynomial of the data, and no statement below proves a new estimate for the Mertens function or the Riemann hypothesis.

Remark 0.2 (Version 8 status addendum). All sections inherited from Versions 1–7 remain part of the notebook. Results later shown to be insufficient are not deleted; they are retained as historical lemmas, experiments, or failed routes. The principal status change is the following:

`PURE LESLIE / ARITHMETIC-FREE ROUTE: CLOSED NEGATIVE`

for the frozen Version 4 predictive protocol described below. The active continuation is an arithmetic-driven, dimension-growing Leslie tower. No novelty claim is made here for the classical divisor identity or for stratification by $\Omega(n)$; the new object under study is their compatibility with the previously developed Leslie geometry.

Remark 0.3 (Version 9 status). Version 9 does not reopen the arithmetic-free predictive route. It retains that negative result and moves the active question one level earlier: arithmetic is supplied exactly by finite divisibility, and the only possible compression is sought in a predeclared observable from that arithmetic state into the Leslie coefficient geometry. The global encoder formulation of Version 8 is retained as a strong realization problem, but the immediate local bridge is the observable-rank problem developed below.

1. Purpose and change of viewpoint

The earlier static experiments asked whether one finite collection of fitted frequencies could be embedded in one nonnegative Leslie matrix. Allowing auxiliary eigenvalues made that static problem feasible in all forty reported configurations, but the result did not distinguish a coherent dynamical law from independent fitting.

The present formulation replaces independent static realizations by a sequence of rigid or fixed-degree repaired Leslie matrices attached to consecutive logarithmic windows. The object of study is therefore a non-autonomous product, or linear cocycle, rather than one fixed matrix. Linear cocycles and their long products are standard objects in dynamical-systems theory [4].

Warning 1.1. The use of a time-dependent matrix product is not itself a new result. The potentially discriminating content is the simultaneous enforcement of rigidity, nonnegativity, continuity through the feasible set, and prediction of a future window without recalibration.

Proposition 1.2 (Three logically distinct layers)**.** *The following problems are distinct and must be tested separately.*

(1) Static spectral completion: *a prescribed critical core polynomial is multiplied by an auxiliary monic factor until a nonnegative Leslie characteristic polynomial is obtained.*
(2) Data-compatible realization: *the characteristic polynomial is the polynomial of an order-D recurrence actually satisfied by the scalar data window.*
(3) Dynamic transport: *two consecutive data-compatible realizations are joined using only the common past equations and a predeclared state rule.*

Static feasibility in the first sense does not imply either of the latter two properties.

Remark 1.3. For the arithmetic programme the logically clean direction is therefore

$$\text{data} \longrightarrow \text{recurrence polynomial} \longrightarrow \text{spectral factorization} \longrightarrow \text{Leslie positivity and transport.}$$

A frequency estimator may remain useful as an exploratory diagnostic, but it is not needed for the exact recurrence-first membership tests developed below.

2. Rigid moving Leslie matrices

Definition 2.1. For $m \geq 2$, a companion Leslie matrix is

$$L(a) = \begin{pmatrix} a_1 & a_2 & \cdots & a_m \\ 1 & 0 & \cdots & 0 \\ 0 & 1 & \ddots & \vdots \\ \vdots & & \ddots & 0 \end{pmatrix}, \qquad a_\ell \geq 0. \tag{2.1}$$

Its characteristic polynomial is

$$P(z; a) = z^m - \sum_{\ell=1}^{m} a_\ell z^{m-\ell}. \tag{2.2}$$

Fix $K \geq 1$ and set

$$m = 2K + 1. \tag{2.3}$$

For $b > 1$ and phases $\Omega = (\omega_1, \ldots, \omega_K)$, prescribe the roots

$$b, \qquad \sqrt{b}\, e^{\pm i\omega_1}, \ldots, \sqrt{b}\, e^{\pm i\omega_K}. \tag{2.4}$$

The unique monic polynomial with these roots is

$$P_{b,\Omega}(z) = (z - b) \prod_{k=1}^{K} \left(z^2 - 2\sqrt{b}\cos(\omega_k) z + b \right). \tag{2.5}$$

Writing

$$P_{b,\Omega}(z) = z^m - \sum_{\ell=1}^{m} a_\ell(b, \Omega) z^{m-\ell} \tag{2.6}$$

defines the coefficient map $A : (b, \Omega) \mapsto a(b, \Omega)$.

Proposition 2.2. *For fixed (b, Ω), the rigid Leslie coefficients in* (2.6) *are unique. A rigid nonnegative Leslie realization exists if and only if*

$$a_\ell(b, \Omega) \geq 0, \qquad 1 \leq \ell \leq m. \tag{2.7}$$

Proof. The roots in (2.4) determine the monic polynomial (2.5). Comparison with (2.2) determines every coefficient. The companion matrix is entrywise nonnegative exactly when its first-row coefficients satisfy (2.7). □

Definition 2.3. The rigid admissible phase set is

$$\mathcal{A}_{b,K} = \left\{ \Omega \in \mathbb{T}^K : a_\ell(b, \Omega) \geq 0 \text{ for all } \ell \right\}. \tag{2.8}$$

Remark 2.4. The boundary of (2.8) is contained in the union of the coefficient zero sets

$$\partial \mathcal{A}_{b,K} \subseteq \bigcup_{\ell=1}^{m} \left\{ \Omega : a_\ell(b, \Omega) = 0 \right\}. \tag{2.9}$$

Equality need not hold without excluding redundant inequalities and singular intersections.

3. Windowed cocycle and accumulated phase

Let $W_0, W_1, \ldots$ be consecutive windows in logarithmic scale. Associate a rigid Leslie matrix

$$L_j = L(a_j), \qquad a_j = a(b_j, \Omega_j) \in \mathbb{R}_+^m. \tag{3.1}$$

The non-autonomous evolution is

$$v_{j+1} = L_j v_j, \tag{3.2}$$

with cocycle product

$$\Phi(J, 0) = L_{J-1} \cdots L_1 L_0. \tag{3.3}$$

Remark 3.1. A moving sequence L_j has no mandatory spectral periodicity. This does not imply that every chosen trajectory is nonperiodic. Periodicity may reappear if the coefficients or phases are selected periodically.

If a locally matched branch has phase $\omega_{j,k}$ in window j, its formal accumulated phase is

$$\Theta_k(J) = \sum_{j=0}^{J-1} \omega_{j,k}. \tag{3.4}$$

The definition requires mode matching and phase unwrapping between adjacent windows. Without those operations, the unordered roots of successive polynomials do not determine a unique trajectory $k \mapsto \omega_{j,k}$.

4. Exact coefficient coordinates for the rigid critical spectrum

The general inverse spectral problem for Leslie matrices has an established characterization in the literature [3]. The results in this section concern the special rigid spectrum in (2.4) and its use as a moving-window coordinate system.

Put $r = \sqrt{b}$ and define

$$S_\Omega(w) = \prod_{k=1}^{K} \left(w^2 - 2\cos(\omega_k)w + 1\right) = \sum_{j=0}^{2K} d_j w^{2K-j}. \tag{4.1}$$

The polynomial is reciprocal, and hence

$$d_{2K-j} = d_j, \qquad d_0 = d_{2K} = 1. \tag{4.2}$$

Moreover,

$$P_{b,\Omega}(z) = (z - r^2) r^{2K} S_\Omega(z/r). \tag{4.3}$$

Proposition 4.1. *The coefficients in* (2.6) *satisfy*

$$a_\ell = r^\ell \left(r d_{\ell-1} - d_\ell\right), \qquad 1 \le \ell \le 2K, \tag{4.4}$$

$$a_m = r^{m+1} = b^{K+1}. \tag{4.5}$$

For $1 \le j \le K$, the paired coefficients are

$$a_j = r^j \left(r d_{j-1} - d_j\right), \tag{4.6}$$

$$a_{m-j} = r^{m-j} \left(r d_j - d_{j-1}\right). \tag{4.7}$$

Proof. Expand (4.3) and compare the coefficient of $z^{m-\ell}$ with (2.6). Equation (4.7) follows from (4.2). □

Corollary 4.2. *A rigid nonnegative Leslie realization exists if and only if*

$$d_j > 0 \quad \textit{and} \quad \frac{1}{r} \le \frac{d_j}{d_{j-1}} \le r, \qquad 1 \le j \le K. \tag{4.8}$$

Strict inequalities are equivalent to strict positivity of all paired coefficients.

Proof. Equations (4.6) and (4.7) give the two inequalities. If one of d_{j-1} and d_j vanished while both paired inequalities held, the other would also vanish. Propagation back to $d_0 = 1$ yields a contradiction. Thus feasibility forces positivity of all d_j for $0 \le j \le K$. □

Definition 4.3. On the region where $d_1, \dots, d_K > 0$, define

$$\eta_j(\Omega) = \log \frac{d_j}{d_{j-1}}, \qquad 1 \le j \le K. \tag{4.9}$$

The rigid spectral strain is

$$\Sigma(\Omega) = \|\eta(\Omega)\|_\infty. \tag{4.10}$$

Corollary 4.4. *The positivity test becomes*

$$\|\eta(\Omega)\|_\infty \le \frac{1}{2}\log b. \tag{4.11}$$

The minimum compatible Perron value is

$$b_*(\Omega) = \exp\left(2\|\eta(\Omega)\|_\infty\right) \tag{4.12}$$

$$= \max_{1\le j\le K}\left\{\left(\frac{d_j}{d_{j-1}}\right)^2, \left(\frac{d_{j-1}}{d_j}\right)^2\right\}. \tag{4.13}$$

If some $d_j \le 0$, set $b_*(\Omega) = +\infty$.

Remark 4.5. If $b > b_*(\Omega)$, the rigid Leslie coefficients are strictly positive. If $b = b_*(\Omega) < \infty$, at least one paired coefficient vanishes. The map b_* is generally nonsmooth where several coordinates attain the maximum in (4.12).

Proposition 4.6. *The rigid root-of-unity comb*

$$\omega_k = \frac{2\pi k}{2K+1}, \qquad 1 \le k \le K, \tag{4.14}$$

has $d_j = 1$ for all j, and hence $\Sigma(\Omega) = 0$. Conversely, $\Sigma(\Omega) = 0$ implies the same multiset of phases, up to ordering and conjugation.

Proof. For (4.14),

$$S_\Omega(w) = \frac{w^{2K+1}-1}{w-1} = 1 + w + \cdots + w^{2K}. \tag{4.15}$$

Conversely, $\eta = 0$ and $d_0 = 1$ imply $d_0 = \cdots = d_{2K} = 1$. The roots of (4.15) are the nontrivial $(2K+1)$-st roots of unity. □

5. Memory tails and the universal critical region

Normalize the coefficients by

$$x_\ell = \frac{a_\ell}{b^\ell}, \qquad X_q = \sum_{\ell=q}^{m} x_\ell. \tag{5.1}$$

Proposition 5.1. *The tails telescope to*

$$X_{j+1} = \frac{d_j}{r^j}, \qquad 0 \le j \le 2K. \tag{5.2}$$

Consequently,

$$X_{m-j} = b^{j-K} X_{j+1}, \qquad 0 \le j \le K, \tag{5.3}$$

$$x_m = X_m = b^{-K}. \tag{5.4}$$

Moreover, positivity is equivalent to

$$\frac{X_j}{b} \le X_{j+1} \le X_j, \qquad 1 \le j \le K. \tag{5.5}$$

Proof. Substitute (4.4) into (5.1); consecutive terms telescope. Equations (5.3)–(5.4) follow from (4.2), and (5.5) is (4.8) rewritten using (5.2). □

The reciprocal polynomial admits a Chebyshev reduction. Define

$$C_n(y) = 2T_n(y/2), \tag{5.6}$$

where T_n is the Chebyshev polynomial of the first kind, and set

$$H_\eta(y) = C_K(y) + d_1 C_{K-1}(y) + \cdots + d_{K-1}C_1(y) + d_K. \tag{5.7}$$

Then

$$S_\Omega(w) = w^K H_\eta(w + w^{-1}). \tag{5.8}$$

Definition 5.2. Let

$$\mathcal{R}_K = \left\{\eta \in \mathbb{R}^K : H_\eta \text{ has } K \text{ simple roots in } (-2,2)\right\}, \tag{5.9}$$

where $d_j = \exp(\eta_1 + \cdots + \eta_j)$.

Proposition 5.3. *In η-coordinates, the admissible rigid critical region is*

$$\mathcal{A}^\eta_{b,K} = \mathcal{R}_K \cap \left[-\frac{1}{2}\log b, \frac{1}{2}\log b\right]^K. \tag{5.10}$$

Thus $\mathcal{R}_K$ encodes the unit-circle root geometry, while the box encodes Leslie positivity.

Remark 5.4. Within the positive-d coordinates, a path can leave (5.10) either through a box face or through the boundary of $\mathcal{R}_K$. The latter occurs when roots collide or reach an endpoint. Since

$$H_\eta(2) = d_K + 2\sum_{j=0}^{K-1} d_j > 0, \tag{5.11}$$

a root cannot reach $+2$. Candidate spectral boundary equations are therefore

$$\operatorname{Disc} H_\eta = 0 \qquad \text{or} \qquad H_\eta(-2) = 0. \tag{5.12}$$

This describes the regular boundary mechanisms; intersections and higher degeneracies form singular strata.

6. Jacobian of the phase-to-strain map

Assume distinct phases $0 < \omega_k < \pi$. Let the phases be ordered to remove permutation ambiguity.

Proposition 6.1. *The coefficient Jacobian satisfies*

$$\left|\det \frac{\partial(d_1,\ldots,d_K)}{\partial(\omega_1,\ldots,\omega_K)}\right| = 2^{K(K+1)/2}\prod_{k=1}^{K}\sin\omega_k \prod_{i<j}|\cos\omega_i - \cos\omega_j|. \tag{6.1}$$

Consequently,

$$\left|\det \frac{\partial\eta}{\partial\Omega}\right| = \frac{2^{K(K+1)/2}\prod_k \sin\omega_k \prod_{i<j}|\cos\omega_i - \cos\omega_j|}{\prod_{j=1}^{K} d_j}. \tag{6.2}$$

Proof. The first K reciprocal coefficients determine the monic polynomial with roots $2\cos\omega_k$ in the Chebyshev basis. The root-to-coefficient Jacobian is a Vandermonde product, the change $2\cos\omega_k$ contributes $2\sin\omega_k$ in absolute value, and the triangular changes between monomial, Chebyshev, and reciprocal coefficient coordinates give the stated power of 2. The map $d \mapsto \eta$ is triangular with determinant $(d_1 \cdots d_K)^{-1}$. □

Corollary 6.2. *The same determinant can be written as*

$$\left|\det \frac{\partial \eta}{\partial \Omega}\right| = \frac{\sqrt{|\mathrm{Disc}\, H_\eta\, H_\eta(2) H_\eta(-2)|}}{d_1 d_2 \cdots d_K}. \tag{6.3}$$

Hence the phase-to-strain map is locally invertible away from phase collisions and the endpoints 0 *and* π*.*

Remark 6.3. At regular points, a moving rigid Leslie model can be transported using K variables η, plus b when the Perron value is allowed to vary. The full coefficient vector is reconstructed from (4.6), (4.7), and (4.5).

7. Exact viability conditions in strain coordinates

For fixed b, the box criterion (4.11) gives the exact first-order viability conditions. On an upper active face,

$$\eta_j = \frac{1}{2}\log b \quad \Longrightarrow \quad \dot\eta_j \le 0, \tag{7.1}$$

and on a lower active face,

$$\eta_j = -\frac{1}{2}\log b \quad \Longrightarrow \quad \dot\eta_j \ge 0. \tag{7.2}$$

Since $\dot\eta = D_\Omega \eta\, \dot\Omega$, both cases are summarized by

$$\mathrm{sgn}(\eta_j) \sum_{k=1}^{K} \frac{\partial \eta_j}{\partial \omega_k} \dot\omega_k \le 0 \tag{7.3}$$

on an active face.

If $b = b(t)$ varies and

$$\eta_j = s_j \frac{1}{2}\log b, \qquad s_j \in \{-1, 1\}, \tag{7.4}$$

then viability requires

$$s_j \dot\eta_j \le \frac{1}{2}\frac{\dot b}{b}. \tag{7.5}$$

For the minimum-Perron choice $b = b_*(\Omega)$ and a unique active index j_*,

$$\frac{\dot b_*}{b_*} = 2\,\mathrm{sgn}(\eta_{j_*})\dot\eta_{j_*}. \tag{7.6}$$

Remark 7.1. When several coordinates attain $\|\eta\|_\infty$, b_* is not differentiable in general. Directional derivatives or a differential inclusion must replace (7.6).

On a fixed upper face with tangential first-order motion, the second-order viability condition is

$$\nabla \eta_j \cdot \ddot\Omega + \dot\Omega^T D^2 \eta_j \dot\Omega \le 0. \tag{7.7}$$

The inequality reverses on a lower face.

8. Exact modal transport and internally generated amplitudes

Assume that every rigid polynomial has distinct roots. For a root λ, define the right eigenvector

$$v(\lambda) = (\lambda^{m-1}, \lambda^{m-2}, \dots, \lambda, 1)^T. \tag{8.1}$$

Let

$$V_j = [v(\lambda_{j,1}) \ \cdots \ v(\lambda_{j,m})], \qquad \Lambda_j = \operatorname{diag}(\lambda_{j,1}, \dots, \lambda_{j,m}). \tag{8.2}$$

If L_j acts for h_j steps and $y_j = V_j c_j$, then

$$y_{j+1} = V_j \Lambda_j^{h_j} c_j, \tag{8.3}$$

$$c_{j+1} = T_j \Lambda_j^{h_j} c_j, \qquad T_j = V_{j+1}^{-1} V_j. \tag{8.4}$$

Proposition 8.1. *The entries of the basis-change matrix are*

$$(T_j)_{pq} = \prod_{\substack{s=1 \\ s \neq p}}^{m} \frac{\lambda_{j,q} - \lambda_{j+1,s}}{\lambda_{j+1,p} - \lambda_{j+1,s}}. \tag{8.5}$$

Proof. The p-th coordinate in the new Vandermonde basis is the cardinal Lagrange polynomial for the nodes $\lambda_{j+1,s}$, evaluated at the old node $\lambda_{j,q}$. Inverse Vandermonde matrices and Lagrange interpolation are equivalent descriptions of this change of basis. □

For a small root displacement

$$\lambda_{j+1,q} = \lambda_{j,q} + \delta\lambda_{j,q}, \tag{8.6}$$

one obtains, for $p \neq q$,

$$(T_j)_{pq} = -\delta\lambda_{j,q} \frac{P_j'(\lambda_{j,q})}{(\lambda_{j,q} - \lambda_{j,p}) P_j'(\lambda_{j,p})} + O(\|\delta\lambda_j\|^2). \tag{8.7}$$

Thus modal mixing becomes large near collisions, consistently with the vanishing of the discriminant.

Proposition 8.2. *The exact modal volume identity is*

$$\det T_j = \frac{\det V_j}{\det V_{j+1}}, \tag{8.8}$$

$$|\det T_j|^2 = \frac{|\operatorname{Disc} P_j|}{|\operatorname{Disc} P_{j+1}|}. \tag{8.9}$$

Proof. The first identity follows from $T_j = V_{j+1}^{-1} V_j$. The square of the absolute Vandermonde determinant equals the absolute discriminant of a monic polynomial with simple roots. □

Iterating (8.4) gives

$$c_J = T_{J-1} \Lambda_{J-1}^{h_{J-1}} \cdots T_0 \Lambda_0^{h_0} c_0, \tag{8.10}$$

and hence

$$y_J = V_J T_{J-1} \Lambda_{J-1}^{h_{J-1}} \cdots T_0 \Lambda_0^{h_0} c_0. \tag{8.11}$$

For the first-coordinate observable,

$$U_J = e_1^T V_J T_{J-1} \Lambda_{J-1}^{h_{J-1}} \cdots T_0 \Lambda_0^{h_0} c_0. \tag{8.12}$$

Warning 8.3. Equations (8.10)–(8.12) eliminate independent amplitude fitting only under the homogeneous state-handoff model (8.3). If each data window is reinitialized, forced, normalized independently, or connected through an additional observation operator, new parameters re-enter and must be counted explicitly.

9. Canonical boundary realization and the remaining law

Whenever $b_*(\Omega) < \infty$, define the canonical minimum-Perron realization

$$L_*(\Omega) = L\big(a(b_*(\Omega), \Omega)\big). \tag{9.1}$$

This realization lies on the boundary of the nonnegative cone. At regular points with unique active strain coordinate, the maps

$$\Omega \longleftrightarrow \eta \longmapsto b_*(\Omega) \longmapsto L_*(\Omega) \tag{9.2}$$

are locally determined, with the first arrow locally invertible.

Remark 9.1. The construction (9.1) is canonical but only piecewise smooth because of the maximum norm in b_*. It is also numerically sensitive near the spectral boundary (5.12).

The static coordinate chain (9.2) does not by itself determine how a future window is reached. The next sections derive an exact overlap law directly from the scalar sequence. For fixed b, the special reciprocal structure turns the admissible coefficient differences into a linear problem. At the first generic stride admitting motion, the entire update is reduced to one scalar innovation.

10. The fixed-Perron reciprocal affine hull

The critical-circle condition has a useful linear precursor. Fix $b > 1$, put $m = 2K + 1$, and for a coefficient vector $a = (a_1, \dots, a_m)$ define the synthetic-division coefficients

$$q_0(a; b) = 1, \tag{10.1}$$

$$q_j(a; b) = b^j - \sum_{\ell=1}^{j} a_\ell b^{j-\ell}, \qquad 1 \le j \le 2K. \tag{10.2}$$

Then

$$P(z; a) = (z - b)\left(z^{2K} + q_1 z^{2K-1} + \cdots + q_{2K}\right) + P(b; a). \tag{10.3}$$

Definition 10.1. The fixed-b reciprocal–Perron affine hull is the set

$$\mathcal{C}^{\mathrm{rec}}_{b,K} = \left\{ a \in \mathbb{R}^m : P(b; a) = 0, \quad q_{2K-j}(a; b) = b^{K-j} q_j(a; b),\ 0 \le j \le K - 1 \right\}. \tag{10.4}$$

Proposition 10.2. *The set* (10.4) *is an affine subspace of dimension K. Equivalently, there is a rank-$(K + 1)$ matrix R_b and a vector c_b such that*

$$\mathcal{C}^{\mathrm{rec}}_{b,K} = \{a : R_b a = c_b\}. \tag{10.5}$$

One explicit choice of the $K + 1$ affine rows is

$$\sum_{\ell=1}^{m} b^{m-\ell} a_\ell = b^m, \tag{10.6}$$

$$\sum_{\ell=1}^{2K-j} b^{2K-j-\ell} a_\ell - \sum_{\ell=1}^{j} b^{K-\ell} a_\ell = b^{2K-j} - b^K, \qquad 0 \le j \le K - 1. \tag{10.7}$$

Every rigid critical Leslie coefficient vector with Perron value b belongs to this affine subspace.

Proof. All equations in (10.4) are affine linear in a. The K quantities $q_1, \dots, q_K$ may be chosen freely. The reciprocal identities determine $q_{K+1}, \dots, q_{2K}$, after which

$$a_j = b q_{j-1} - q_j, \qquad 1 \le j \le 2K, \tag{10.8}$$

and $P(b; a) = 0$ determines

$$a_m = b q_{2K} = b^{K+1}. \tag{10.9}$$

Hence the affine dimension is K. For a rigid critical spectrum, $q_j = r^j d_j$ with $r = \sqrt{b}$, and the palindromy $d_{2K-j} = d_j$ gives exactly the reciprocal equations in (10.4). □

Remark 10.3. Membership in $\mathcal{C}^{\mathrm{rec}}_{b,K}$ is necessary but not sufficient for all non-Perron roots to lie on $|z| = \sqrt{b}$. Put

$$d_j = \frac{q_j}{b^{j/2}}. \tag{10.10}$$

The reciprocal identities give $d_{2K-j} = d_j$. Full criticality additionally requires the associated polynomial H in (5.7) to have all K roots in $[-2, 2]$, together with the Leslie nonnegativity inequalities.

11. Overlapping windows and exact mobility rigidity

Let (u_t) be any scalar sequence. In the intended arithmetic application one may take $u_t = M(B_t)$ or a predeclared normalization of those values. Define the recurrence row

$$h_t = (u_{t-1}, u_{t-2}, \dots, u_{t-m}) \in \mathbb{R}^m. \tag{11.1}$$

The order-m recurrence window beginning at n consists of the m equations

$$h_t a^{(n)} = u_t, \qquad n + m \le t \le n + 2m - 1. \tag{11.2}$$

When the corresponding square data matrix is nonsingular, these equations determine $a^{(n)}$ uniquely.

Let a second window begin at $n + s$, with $1 \le s \le m$. The two recurrence windows share exactly $m - s$ equations. Let $O_{n,s}$ be the matrix whose rows are the common h_t:

$$O_{n,s} = \begin{pmatrix} h_{n+s+m} \\ \vdots \\ h_{n+2m-1} \end{pmatrix} \in \mathbb{R}^{(m-s)\times m}, \tag{11.3}$$

with the empty-matrix convention when $s = m$.

Theorem 11.1 (Exact overlap constraint)**.** *Assume that $a^{(n)}$ and $a^{(n+s)}$ both fit their respective windows and both belong to $\mathcal{C}^{\mathrm{rec}}_{b,K}$ for the same fixed b. Then*

$$\delta a_{n,s} := a^{(n+s)} - a^{(n)} \in \ker O_{n,s} \cap \ker R_b. \tag{11.4}$$

Equivalently, with

$$Z_{n,s} = \begin{pmatrix} O_{n,s} \\ R_b \end{pmatrix}, \tag{11.5}$$

one has

$$Z_{n,s}\delta a_{n,s} = 0. \tag{11.6}$$

Proof. Every common recurrence equation is satisfied by both coefficient vectors, so subtraction gives $O_{n,s}\delta a_{n,s} = 0$. Since both vectors satisfy the same affine equations $R_b a = c_b$, subtraction also gives $R_b \delta a_{n,s} = 0$. □

Corollary 11.2 (Generic mobility threshold)**.** *If the rows of $O_{n,s}$ and R_b are in generic relative position, then*

$$\dim \ker Z_{n,s} = \max\{0, s - K - 1\}. \tag{11.7}$$

Hence:

$$s \le K + 1 \implies \text{the old coefficient vector is locally isolated}, \tag{11.8}$$

$$s = K + 2 \implies \text{one generic coefficient direction survives}, \tag{11.9}$$

$$s > K + 2 \implies s - K - 1 \text{ generic directions survive}. \tag{11.10}$$

Proof. The common-equation matrix contributes $m-s$ generic independent rows and R_b contributes $K+1$ independent rows. Since $m = 2K+1$, the generic nullity is

$$m - \min\{m, m-s+K+1\} = \max\{0, s-K-1\}. \tag{11.11}$$

□

Definition 11.3. For fixed b, define the critical stride

$$s_* = K+2. \tag{11.12}$$

It is the first stride at which a nonzero fixed-b reciprocal–Perron motion exists generically.

Remark 11.4. For $s \le K+1$, nontrivial motion is possible only when $Z_{n,s}$ loses rank. Thus exceptional motion is detected by the vanishing of maximal minors of $Z_{n,s}$. In particular, any single Hankel-type minor obtained from the fixed identity $a_m = b^{K+1}$ is only one necessary degeneracy condition, not a sufficient condition for full critical motion.

Remark 11.5. If b is allowed to vary, the regular rigid critical manifold has dimension $K+1$. A transversality count gives local intersection dimension

$$\max\{0, s-K\}, \tag{11.13}$$

so the first generic one-dimensional mobility occurs at $s = K+1$. Unlike the fixed-b result, this statement is local and nonlinear because coefficient differences for distinct Perron values do not lie in one fixed affine subspace.

12. Scalar innovation at the critical stride

Set $s = s_* = K+2$ and assume

$$\operatorname{rank} Z_{n,s_*} = m-1. \tag{12.1}$$

Choose a nonzero vector v_n spanning the one-dimensional nullspace,

$$\ker Z_{n,s_*} = \operatorname{span}\{v_n\}. \tag{12.2}$$

Theorem 12.1 (Scalarization of the window update)**.** *Under* (12.1)*, every next fixed-b reciprocal–Perron coefficient vector that agrees with the old window on all common equations has the form*

$$a^{(n+s_*)} = a^{(n)} + \tau_n v_n \tag{12.3}$$

for a unique scalar τ_n*.*

Proof. The difference belongs to the one-dimensional nullspace in (12.2). □

The next window contributes $s_* = K+2$ genuinely new recurrence equations. The first occurs at

$$t_0 = n+2m. \tag{12.4}$$

Its recurrence row h_{t_0} uses only values already known before u_{t_0} is observed.

Theorem 12.2 (One innovation sample determines the next Leslie)**.** *Assume*

$$h_{t_0} v_n \neq 0. \tag{12.5}$$

Then the first new scalar value determines the update parameter exactly:

$$\boxed{\tau_n = \frac{u_{t_0} - h_{t_0} a^{(n)}}{h_{t_0} v_n}}. \tag{12.6}$$

Consequently the entire next coefficient vector is fixed by (12.3)*.*

Proof. Substitute (12.3) into the recurrence equation at t_0 and solve the resulting scalar equation for τ_n. □

Corollary 12.3 (Causal $K+1$-value forecast)**.** *After observing only u_{t_0}, define $a^+ = a^{(n)} + \tau_n v_n$ using* (12.6)*. Then recursively define*

$$\widehat{u}_t = \sum_{\ell=1}^{m} a_\ell^+ \widehat{u}_{t-\ell}, \qquad t_0+1 \le t \le t_0+K+1, \tag{12.7}$$

using the observed historical values for indices below t_0 and earlier forecasts above t_0. If the next window is exactly generated by a fixed-b rigid critical Leslie coefficient vector, then

$$\widehat{u}_t = u_t, \qquad t_0+1 \le t \le t_0+K+1. \tag{12.8}$$

Proof. The first new observation determines the exact next coefficient vector. The remaining $K+1$ new equations of that window use the same coefficient vector. Induction on t in (12.7) gives the result. □

Proposition 12.4 (Rank-one block diagnostic)**.** *Let G_n contain the $K+2$ new recurrence rows of the next window, evaluated from the observed sequence, and let y_n^+ be their right-hand side. Define*

$$e_n = y_n^+ - G_n a^{(n)}, \qquad g_n = G_n v_n. \tag{12.9}$$

Then exact critical-stride transport implies

$$e_n = \tau_n g_n. \tag{12.10}$$

Equivalently,

$$(e_n)_i (g_n)_j - (e_n)_j (g_n)_i = 0 \qquad \text{for all } i,j. \tag{12.11}$$

Thus a block of $K+2$ new equations carries only one scalar innovation after the old state and overlap direction are fixed.

13. Admissible innovation interval and boundary locking

The scalar parameter in (12.3) cannot take arbitrary values if nonnegativity and the critical-circle root condition are retained.

Definition 13.1. For a current nonnegative coefficient vector a and direction v, define the interval of coefficient positivity

$$I_{\mathrm{pos}}(a,v) = \left[\max_{v_\ell>0} \left(-\frac{a_\ell}{v_\ell}\right),\ \min_{v_\ell<0} \left(-\frac{a_\ell}{v_\ell}\right) \right], \tag{13.1}$$

with the usual $-\infty$ and $+\infty$ conventions when one of the indexing sets is empty.

For $a(\tau) = a + \tau v$, define $q_j(\tau)$ by (10.2), $d_j(\tau) = q_j(\tau)/b^{j/2}$, and the corresponding reciprocal reduction H_τ. Let I_{spec} be the connected component containing 0 of the set of τ for which H_τ has K simple roots in $(-2,2)$. At regular points, an endpoint of this component can occur only when

$$\operatorname{Disc} H_\tau = 0 \qquad \text{or} \qquad H_\tau(-2) = 0. \tag{13.2}$$

Define the full local innovation interval

$$I_{\mathrm{adm}} = I_{\mathrm{pos}} \cap I_{\mathrm{spec}}. \tag{13.3}$$

Corollary 13.2 (Pre-observation admissible range)**.** *Before u_{t_0} is observed, the moving rigid Leslie hypothesis requires*

$$\boxed{u_{t_0} \in h_{t_0} a^{(n)} + (h_{t_0} v_n) I_{\mathrm{adm}}.} \tag{13.4}$$

A value outside this interval falsifies the local fixed-b critical transition before any later value of the new block is used.

Proposition 13.3 (Boundary locking criterion)**.** *Let*

$$\mathcal{Z}(a) = \{\ell : a_\ell = 0\} \tag{13.5}$$

be the active coefficient faces. A nonzero one-sided coefficient motion $a+\tau v$ preserving nonnegativity exists locally only if the numbers v_ℓ, $\ell \in \mathcal{Z}(a)$, do not contain both positive and negative values. If both signs occur, then $\tau = 0$ is the only sufficiently small feasible update.

Proof. For each active coordinate the inequality $a_\ell + \tau v_\ell \geq 0$ reduces to $\tau v_\ell \geq 0$. A common nonzero sign of τ exists exactly when the active v_ℓ do not impose conflicting signs. □

Remark 13.4. The canonical minimum-Perron realization $L_*(\Omega)$ always has at least one active coefficient face. It is therefore dynamically more restrictive than an interior choice $b > b_*(\Omega)$. A canonical boundary trajectory may lock whenever the critical-stride direction intersects active faces with incompatible signs.

14. Exact scalar-driven strain transport

For the update $a^+ = a + \tau v$, define

$$\psi_j(v; b) = \sum_{\ell=1}^{j} v_\ell b^{j-\ell}, \qquad \psi_0 = 0. \tag{14.1}$$

Equation (10.2) gives

$$q_j(a + \tau v; b) = q_j(a; b) - \tau \psi_j(v; b). \tag{14.2}$$

Theorem 14.1 (Exact strain update)**.** *Whenever all logarithms are defined,*

$$\boxed{\eta_j^+ - \eta_j = \log\left(1 - \tau \frac{\psi_j}{q_j}\right) - \log\left(1 - \tau \frac{\psi_{j-1}}{q_{j-1}}\right), \qquad 1 \leq j \leq K.} \tag{14.3}$$

Hence, at the critical stride, all K strain coordinates are driven by the same scalar innovation τ.

Proof. Use $d_j = q_j / b^{j/2}$ in the definition $\eta_j = \log(d_j/d_{j-1})$ and substitute (14.2). □

Differentiating with respect to the scalar innovation gives the exact line-flow equation

$$\frac{d\eta_j}{d\tau} = -\frac{\psi_j}{q_j - \tau\psi_j} + \frac{\psi_{j-1}}{q_{j-1} - \tau\psi_{j-1}}. \tag{14.4}$$

For small τ,

$$\eta_j^+ - \eta_j = -\tau\left(\frac{\psi_j}{q_j} - \frac{\psi_{j-1}}{q_{j-1}}\right) + O(\tau^2). \tag{14.5}$$

Thus the K-dimensional strain increment is generically confined to one first-order direction at the critical stride.

15. Frequency-free determinant identities

The reciprocal–Perron test can be imposed directly on a data window without first estimating any phases. Let

$$\mathsf{A}_n = \begin{pmatrix} h_{n+m} \\ \vdots \\ h_{n+2m-1} \end{pmatrix}, \qquad \mathsf{g}_n = \begin{pmatrix} u_{n+m} \\ \vdots \\ u_{n+2m-1} \end{pmatrix}. \tag{15.1}$$

Write the affine reciprocal constraints in (10.5) row by row as

$$\rho_\alpha(b)^T a = \kappa_\alpha(b), \qquad 0 \leq \alpha \leq K. \tag{15.2}$$

For each row define

$$\Delta_{\alpha,n}(b) = \det \begin{pmatrix} \mathsf{A}_n & \mathsf{g}_n \\ \rho_\alpha(b)^T & \kappa_\alpha(b) \end{pmatrix}, \tag{15.3}$$

Proposition 15.1. *If $\det \mathsf{A}_n \neq 0$, then the unique order-m recurrence coefficient vector of the window belongs to $\mathcal{C}^{\mathrm{rec}}_{b,K}$ if and only if*

$$\Delta_{\alpha,n}(b) = 0, \qquad 0 \le \alpha \le K. \tag{15.4}$$

Full rigid criticality is obtained by adding the nonnegativity and root-location tests of Sections 4–5.

Proof. Since $a = \mathsf{A}_n^{-1}\mathsf{g}_n$, the Schur-complement identity gives

$$\Delta_{\alpha,n}(b) = \det(\mathsf{A}_n)\left(\kappa_\alpha(b) - \rho_\alpha(b)^T \mathsf{A}_n^{-1}\mathsf{g}_n\right). \tag{15.5}$$

Thus the determinants vanish exactly when all affine reciprocal–Perron constraints hold. □

Remark 15.2. If b is not fixed in advance, the functions $\Delta_{\alpha,n}(b)$ are explicit polynomials in b whose coefficients depend only on the data window. A necessary condition for a rigid critical realization with some $b > 1$ is that these $K+1$ polynomials possess a common real root $b > 1$. This can be tested without frequencies by polynomial gcd or resultant/subresultant elimination, followed by the root-location and positivity conditions.

16. Closed form for one mode pair

For $K = 1$ one has $m = 3$, and the entire construction becomes explicit. Write $\alpha = a_1$. The fixed-b reciprocal–Perron constraints give

$$\boxed{a_1 = \alpha, \qquad a_2 = b^2 - b - b\alpha, \qquad a_3 = b^2.} \tag{16.1}$$

The quotient polynomial is

$$Q(z) = z^2 + (b - \alpha)z + b. \tag{16.2}$$

Its two roots lie on $|z| = \sqrt{b}$, and the Leslie coefficients are nonnegative, if and only if

$$\boxed{\max\{0, b - 2\sqrt{b}\} \le \alpha \le b - 1.} \tag{16.3}$$

Endpoint equalities include coefficient or spectral degeneracies; strict interior conditions give the corresponding nondegenerate case.

Define

$$X_t(b) = u_{t-1} - bu_{t-2}, \tag{16.4}$$

$$Y_t(b) = u_t - (b^2 - b)u_{t-2} - b^2 u_{t-3}. \tag{16.5}$$

Then the critical recurrence is simply

$$Y_t(b) = \alpha X_t(b). \tag{16.6}$$

Thus, if (X_t, X_{t+1}, X_{t+2}) is not identically zero, three consecutive recurrence equations admit a common fixed-b reciprocal coefficient vector if and only if all pairwise minors vanish:

$$X_i Y_j - X_j Y_i = 0, \qquad t \le i < j \le t + 2. \tag{16.7}$$

If $X_t = X_{t+1} = X_{t+2} = 0$, existence instead requires $Y_t = Y_{t+1} = Y_{t+2} = 0$. Full rigid criticality additionally requires the inferred α to satisfy (16.3).

For $K = 1$, the critical stride is $s_* = 3$ and the fixed-b affine direction is

$$v = (1, -b, 0)^T. \tag{16.8}$$

Hence

$$\alpha^+ = \alpha + \tau, \qquad a_2^+ = a_2 - b\tau, \qquad a_3^+ = b^2. \tag{16.9}$$

The first new recurrence equation gives

$$\tau = \frac{u_{t_0} - h_{t_0} a}{X_{t_0}(b)}, \tag{16.10}$$

provided $X_{t_0}(b) \neq 0$, and the next two values are then fixed recursively. The exact admissible scalar interval is

$$\boxed{\max\{0, b - 2\sqrt{b}\} - \alpha \leq \tau \leq b - 1 - \alpha.} \tag{16.11}$$

This gives a completely frequency-free three-step falsification and two-value forecasting rule for the one-pair model.

17. Irreducible scalar forcing and the endpoint of the Leslie geometry

Proposition 17.1. *At a regular interior fixed-b point and the critical stride $s_* = K + 2$, the Leslie geometry cannot determine τ_n from the old window and the common equations alone. In a neighborhood of $\tau = 0$, distinct values of τ produce distinct reciprocal–Perron coefficient vectors satisfying all those old and common constraints. Full criticality persists for a nontrivial interval whenever the current point lies in the interior of the fixed-b critical region.*

Proof. The rank assumption (12.1) leaves exactly the affine line (12.3). Within the reciprocal affine hull, coefficient positivity and the condition that the reduced polynomial have simple roots in $(-2, 2)$ are open. Hence sufficiently small values of τ remain admissible. □

Remark 17.2. This identifies the exact limit of the fixed-b rigid moving-Leslie mechanism. The rigid geometry compresses a block of $K + 2$ genuinely new recurrence equations to one scalar innovation and then transports all K strain coordinates through (14.3). It does not supply a law for that scalar innovation. A zero-latency forecast of the first new value therefore requires an additional arithmetic statement about the sequence τ_n. Any such law is external to Leslie geometry and is the remaining place where genuinely Möbius-specific information would have to enter.

18. Exact first-order spectral transport

Let $a(t)$ be a differentiable coefficient path and define

$$P(z, t) = z^m - \sum_{\ell=1}^{m} a_\ell(t) z^{m-\ell}. \tag{18.1}$$

Let $\lambda_k(t)$ be a simple root, so that

$$P(\lambda_k(t), t) = 0, \qquad P_z(\lambda_k(t), t) \neq 0. \tag{18.2}$$

Differentiability of simple eigenvalue branches is standard in matrix perturbation theory [5].

Proposition 18.1. *Under* (18.2),

$$\dot{\lambda}_k = \frac{\displaystyle\sum_{\ell=1}^{m} \dot{a}_\ell \lambda_k^{m-\ell}}{P_z(\lambda_k, t)}. \tag{18.3}$$

Proof. Differentiating $P(\lambda_k(t), t) = 0$ gives

$$P_z(\lambda_k, t)\dot{\lambda}_k + P_t(\lambda_k, t) = 0. \tag{18.4}$$

Since

$$P_t(z, t) = -\sum_{\ell=1}^{m} \dot{a}_\ell z^{m-\ell}, \tag{18.5}$$

substitution into (18.4) yields (18.3). □

Suppose now that

$$\lambda_k(t) = \sqrt{b(t)}e^{i\omega_k(t)}. \tag{18.6}$$

Then

$$\frac{\dot\lambda_k}{\lambda_k} = \frac{\dot b}{2b} + i\dot\omega_k. \tag{18.7}$$

Define

$$J_{k\ell}(L) = \frac{\lambda_k^{m-\ell-1}}{P_z(\lambda_k)}. \tag{18.8}$$

Corollary 18.2. *For every simple prescribed branch,*

$$\dot\omega_k = \Im\left(\sum_{\ell=1}^{m} J_{k\ell}(L)\dot a_\ell\right), \tag{18.9}$$

$$\frac{\dot b}{2b} = \Re\left(\sum_{\ell=1}^{m} J_{k\ell}(L)\dot a_\ell\right). \tag{18.10}$$

Warning 18.3. Equation (18.10) must give the same scalar $\dot b/(2b)$ for every k. Thus the imaginary-part formula alone is not a complete admissibility test. The real-part compatibility equations are additional local constraints.

19. The rigid critical manifold

Let

$$\mathcal{M}_K = \left\{a(b,\Omega) : b > 1,\ \Omega \in \mathbb{T}^K,\ a(b,\Omega) \geq 0\right\} \subseteq \mathbb{R}^{2K+1}. \tag{19.1}$$

Proposition 19.1. *At a point where the coefficient map $(b,\Omega) \mapsto a(b,\Omega)$ has rank $K+1$ and the prescribed roots are distinct, $\mathcal{M}_K$ is locally a smooth manifold of dimension $K+1$ and codimension K in $\mathbb{R}^{2K+1}$.*

Proof. The parameter space has dimension $K+1$. Under the stated full-rank hypothesis, the constant-rank theorem gives a local image manifold of dimension $K+1$. The ambient dimension is $2K+1$, so the codimension is K. □

Consequently, locally there are K independent constraints

$$F_r(a) = 0, \qquad 1 \leq r \leq K, \tag{19.2}$$

and every differentiable path in the regular part satisfies

$$\nabla F_r(a) \cdot \dot a = 0, \qquad 1 \leq r \leq K. \tag{19.3}$$

Warning 19.2. The dimension statement is generic, not universal. Repeated roots, collisions of phases, loss of rank, and intersections with several active coefficient faces can create singular or lower-dimensional strata.

20. Positivity and the corrected spectral velocity cone

For $a \in \mathbb{R}^m_+$, the tangent cone of the nonnegative orthant is

$$T_a\mathbb{R}^m_+ = \{v \in \mathbb{R}^m : v_\ell \geq 0 \text{ whenever } a_\ell = 0\}. \tag{20.1}$$

Tangent cones generalize tangent spaces at nonsmooth boundary points of convex sets [6].

The velocity must satisfy three types of conditions simultaneously:

(1) orthant feasibility, $v \in T_a\mathbb{R}^m_+$;
(2) tangency to the rigid critical manifold, $DF(a)v = 0$;

(3) common radial motion across all prescribed root pairs.

Define the common-radial constraint operator by

$$\mathcal{R}_L v = \left(\Re(J_{2\cdot}(L)v) - \Re(J_{1\cdot}(L)v), \dots, \Re(J_{K\cdot}(L)v) - \Re(J_{1\cdot}(L)v)\right)^T. \tag{20.2}$$

At a regular point, define the admissible coefficient-velocity cone

$$\mathcal{K}(L) = T_a\mathbb{R}^m_+ \cap \ker DF(a) \cap \ker \mathcal{R}_L. \tag{20.3}$$

Definition 20.1. The rigid positive spectral velocity cone is

$$\mathcal{V}(L) = \{\Im J(L)v : v \in \mathcal{K}(L)\}. \tag{20.4}$$

Proposition 20.2. *Every differentiable rigid positive Leslie path with simple prescribed roots satisfies*

$$\dot{\Omega}(t) \in \mathcal{V}(L(t)). \tag{20.5}$$

Proof. The coefficient velocity lies in the orthant tangent cone, is tangent to the rigid critical manifold, and satisfies the common-radial identities. Equation (18.9) then places its phase velocity in the image (20.4). □

Remark 20.3. Condition (20.5) is first-order and necessary. At singular points or for finite window increments it is not sufficient for the existence of a global path.

21. Second-order transport

Proposition 21.1. *If $a(t)$ is twice differentiable and* (18.2) *holds, then*

$$\ddot{\lambda}_k = -\frac{P_{zz}\dot{\lambda}_k^2 + 2P_{zt}\dot{\lambda}_k + P_{tt}}{P_z}, \tag{21.1}$$

where all polynomial derivatives are evaluated at $(\lambda_k(t), t)$ and

$$P_{tt}(z,t) = -\sum_{\ell=1}^{m} \ddot{a}_\ell z^{m-\ell}. \tag{21.2}$$

Proof. Differentiate (18.4) once more and collect the chain-rule terms. Division by $P_z \neq 0$ gives (21.1). □

For the parameterization (18.6),

$$\frac{\ddot{\lambda}_k}{\lambda_k} = \frac{\ddot{b}}{2b} - \frac{\dot{b}^2}{4b^2} - \dot{\omega}_k^2 + i\left(\ddot{\omega}_k + \frac{\dot{b}}{b}\dot{\omega}_k\right). \tag{21.3}$$

Thus a smooth Leslie path constrains not only Ω and $\dot{\Omega}$, but also the angular acceleration after the radial terms are removed.

Definition 21.2. At a regular path point, the residual obtained by comparing the observed $\ddot{\Omega}$ with the imaginary part implied by (21.1) is called the local Leslie spectral-curvature residual.

Warning 21.3. A numerical curvature residual depends on the chosen interpolation of discrete windows and on the time scale assigned to the window index. It is not intrinsic until those choices are fixed before testing.

22. Discrete windows and finite increments

The observations are discrete, not a differentiable curve. Let

$$\Delta a_j = a_{j+1} - a_j, \qquad \Delta\Omega_j = \operatorname{unwrap}(\Omega_{j+1} - \Omega_j). \tag{22.1}$$

The differential model suggests the residual

$$r_j = \Delta\Omega_j - \Im J(L_j)\Delta a_j. \tag{22.2}$$

Warning 22.1. Equation (22.2) is only a first-order approximation. A valid finite-step test should also check that the segment or reconstructed path remains in the admissible set and should report an error scale of order $\|\Delta a_j\|^2$ away from singularities.

A stronger discrete feasibility test is to solve for a path $a_j(s)$, $0 \le s \le 1$, satisfying

$$a_j(0) = a_j, \qquad a_j(1) = a_{j+1}, \qquad a_j(s) \in \mathcal{M}_K \cap \mathbb{R}^m_+. \tag{22.3}$$

Existence of (22.3) is stronger than endpoint feasibility.

23. A genuine forecasting protocol

Experiment 23.1. Choose before examining the test windows:

(1) a fixed base b or a predeclared rule for b_j;
(2) a fixed number K of mode pairs and hence $m = 2K + 1$;
(3) consecutive, nonoverlapping or explicitly overlapping logarithmic windows;
(4) a frequency estimator and a deterministic mode-matching rule;
(5) a coefficient-transport law trained only on past windows;
(6) a rule for handling windows outside $\mathcal{A}_{b,K}$;
(7) a horizon and evaluation metric.

For each training endpoint J, infer $L_0, \dots, L_J$, predict $\widehat{a}_{J+1}$ without using W_{J+1}, project to the admissible rigid set if that projection was predeclared, compute $\widehat{\Omega}_{J+1}$ from the roots, and only then evaluate the prediction on W_{J+1}.

The information flow must be

$$M|_{W_0,\dots,W_J} \longrightarrow L_0, \dots, L_J \longrightarrow \widehat{L}_{J+1} \longrightarrow \widehat{\Omega}_{J+1} \longrightarrow \widehat{M}|_{W_{J+1}}. \tag{23.1}$$

Warning 23.2. The identities of Sections 18–21 do not themselves specify a forecasting law for $\dot{a}$ or $\ddot{a}$. A predictor requires a predeclared dynamical model for coefficient evolution. Merely estimating L_{J+1} from the future window would return to independent fitting.

24. Critical-stride forecasting protocol

Experiment 24.1. Fix K, $m = 2K + 1$, a Perron value $b > 1$, and the critical stride $s_* = K + 2$ before examining the test region. For each training window:

(1) solve the m recurrence equations (11.2) directly from the past scalar sequence;
(2) reject the window unless the resulting coefficients satisfy the fixed-b reciprocal determinants (15.4), nonnegativity, and the critical-circle root-location test;
(3) construct Z_{n,s_*} using only common equations and R_b;
(4) require $\operatorname{rank} Z_{n,s_*} = m - 1$ and compute its null direction v_n;
(5) compute the pre-observation interval (13.4);
(6) reveal only the first new scalar u_{t_0} and compute τ_n from (12.6);
(7) without recalibration, predict the remaining $K + 1$ new values using (12.7);
(8) record the $K + 1$ forecast errors, the scalar innovation τ_n, the strain update (14.3), and whether $\tau_n \in I_{\mathrm{adm}}$.

The information flow in this strongest fixed-b test is therefore

$$\text{past block} \longrightarrow (a^{(n)}, v_n, I_{\mathrm{adm}}) \longrightarrow u_{t_0} \longrightarrow \tau_n \longrightarrow \widehat{u}_{t_0+1}, \dots, \widehat{u}_{t_0+K+1}, \tag{24.1}$$

where u_{t_0} is the single newly revealed scalar value; no later test value is used to update the coefficients.

Remark 24.2. If the $K+1$-value forecasts succeed repeatedly, the remaining arithmetic object is the scalar innovation sequence (τ_n). A further predictive law for τ_n would upgrade the one-sample-delayed forecast to a genuine zero-latency block forecast. Leslie geometry alone does not provide such a law by Proposition 17.1.

25. Four nested models

Definition 25.1. The comparison should separate:

Model A. free frequencies or unconstrained recurrences fitted independently in every window;
Model B. rigid nonnegative Leslie matrices fitted independently in every window;
Model C. one dynamically transported rigid nonnegative Leslie trajectory, forecast from past windows only;
Model D. one dynamically transported *data-compatible repaired* Leslie trajectory with fixed K, fixed auxiliary degree q, and fixed or predeclared Perron rule.

Model B tests static rigid realizability. Model C tests rigid transport. Model D is the appropriate extension when auxiliary roots are genuinely necessary, but it is admissible only after the repaired overlap law has been verified; independently minimizing an auxiliary completion in each window is not Model D. Models C and D should be compared not only with Models A and B, but also with persistence, autoregressive, low-rank Hankel, phase-randomized, support-preserving, and multiplicative null models.

26. What the previous forty runs do and do not establish

Experiment 26.1. The reported forty runs varied the sampling base, training fraction, and requested number of modes. They established numerical out-of-sample prediction by free frequencies and static positive spectral completion with auxiliary eigenvalues for those forty configurations.

Proposition 26.2. *The forty configurations do not constitute a temporal Leslie trajectory and cannot test* (20.5) *or the forecast protocol* (23.1)*.*

Proof. A temporal test requires consecutively ordered windows under fixed or predeclared modelling choices. The forty runs index alternative parameter configurations rather than successive time windows of one fixed model. □

27. Verified corrections to the proposed formulation

Proposition 27.1. *The first derivative* (18.3)*, the angular and radial decompositions* (18.9)–(18.10)*, and the second derivative* (21.1) *are algebraically correct under the simple-root hypothesis.*

Remark 27.2. An independent finite-difference calculation was performed for a degree-five companion polynomial with randomly chosen coefficient velocities. The maximum discrepancy between the finite-difference root velocity and (18.3) was below 9×10^{-9} for a step of 10^{-7}. This is a numerical consistency check, not a substitute for Proposition 18.1.

Remark 27.3. Three corrections are essential:

(1) the boundary relation is the inclusion (2.9), not automatic equality;
(2) the spectral velocity cone must include tangency to the rigid manifold and common radial compatibility, as in (20.3);
(3) the dimension and codimension counts are generic and require a full-rank hypothesis.

28. Numerical obstruction for the rigid $K = 7$ core

The rigid feasibility criterion of Corollary 4.2 can be evaluated even when the logarithmic coordinates η_j do not exist. For $r = \sqrt{b}$ define

$$\delta_j(b, \Omega) = \min\left\{rd_{j-1} - d_j, rd_j - d_{j-1}\right\}, \qquad 1 \le j \le K, \tag{28.1}$$

and

$$\Delta_K(b, \Omega) = \min_{1 \le j \le K} \delta_j(b, \Omega). \tag{28.2}$$

Because the factors multiplying the two quantities in (28.1) in the paired Leslie coefficients are positive, one has

(28.3)

$$\Delta_K(b, \Omega) \ge 0 \quad \Longleftrightarrow \quad P_{b,\Omega} \text{ is the characteristic polynomial of a nonnegative rigid Leslie matrix.}$$

Unlike $b_*(\Omega)$, the margin (28.2) remains defined when one or more d_j are nonpositive.

Experiment 28.1. A grid of 241 values of b in $[1.01, 1.25]$ was examined with $K = 7$ and the predeclared frequency estimator. The estimator returned all seven frequencies at eight grid points. For every one of these eight evaluable configurations,

$$\Delta_7(b, \Omega) < 0. \tag{28.4}$$

The largest observed margin was

$$\boxed{\Delta_{\max} = -1.228610846762} \qquad \text{at} \qquad \boxed{b = 1.178}. \tag{28.5}$$

Thus none of the eight evaluable critical spectra is the complete spectrum of a nonnegative Leslie matrix of rigid dimension 15.

Seven of the eight configurations had $d_1, \ldots, d_7 > 0$ and therefore admitted the logarithmic coordinates η and the finite quantity $b_*(\Omega)$. In all seven cases $b_*(\Omega) > b$. The eighth configuration had all seven frequencies but at least one nonpositive d_j, so b_* was not finite although Δ_7 remained evaluable.

The index attaining the most negative paired margin had the empirical distribution

j_*	2	3	4	5	6
count	1	1	3	1	2.

(28.6)

In particular, $j_* = 1$ did not occur. The dominant obstruction in these eight cases is therefore not the first hemispheric condition alone; it occurs in higher symmetric combinations of the seven phases.

Remark 28.2. Experiment 28.1 is a statement about the eight grid points at which the full $K = 7$ core was available. The other 233 points are not evaluations of Δ_7 and must not be counted as rigidly feasible or infeasible. Likewise, a reported prefix statistic $\kappa = 0$ only excludes the tested angularly ordered prefixes; it does not exclude every possible subfamily of the seven frequencies.

Corollary 28.3. *For each configuration in Experiment 28.1 for which a nonnegative Leslie realization containing the same prescribed roots exists in a larger dimension, at least one auxiliary root is necessary.*

Proof. Equation (28.4) excludes degree-zero completion, i.e. the rigid characteristic polynomial itself. Any larger realization containing the same roots therefore has a nonconstant complementary factor. □

29. Minimal auxiliary completion as an exact linear feasibility problem

The negative rigid test does not determine how many auxiliary eigenvalues are necessary. That question has an exact finite-dimensional formulation that does not require re-estimating any frequency.

Let

$$C(z) = z^d + c_1 z^{d-1} + \cdots + c_d, \qquad d = 2K+1, \tag{29.1}$$

be a fixed rigid core polynomial $P_{b,\Omega}$, so $C(b) = 0$. For $q \geq 0$ let

$$R_q(z) = z^q + r_1 z^{q-1} + \cdots + r_q \tag{29.2}$$

be a real monic polynomial and write

$$C(z)R_q(z) = z^{d+q} + \sum_{k=1}^{d+q} p_k z^{d+q-k}. \tag{29.3}$$

Definition 29.1. The auxiliary completion index of the core C is

$$\nu(C) = \min\left\{q \geq 0 : \exists R_q \text{ monic real with } p_k \leq 0 \text{ for every } k \geq 1\right\}, \tag{29.4}$$

with $\nu(C) = +\infty$ if no such polynomial exists. For $C = P_{b,\Omega}$ we also write $\nu_b(\Omega) = \nu(C)$.

Proposition 29.2. *A degree-q auxiliary completion in Definition 29.1 exists if and only if there is a nonnegative Leslie companion matrix of dimension $d+q$ whose spectrum contains all roots of C and whose Perron eigenvalue is b.*

Proof. If $p_k \leq 0$, set $a_k = -p_k \geq 0$. Then (29.3) is

$$z^{d+q} - \sum_{k=1}^{d+q} a_k z^{d+q-k},$$

the characteristic polynomial of a nonnegative Leslie companion matrix. Since $C(b) = 0$, the product also vanishes at b. The vector

$$(b^{d+q-1}, b^{d+q-2}, \ldots, b, 1)^T$$

is strictly positive and is an eigenvector with eigenvalue b. The Collatz–Wielandt characterization therefore gives spectral radius b.

Conversely, if a nonnegative Leslie characteristic polynomial contains all roots of C, polynomial division by the monic factor C gives a real monic complementary polynomial R_q, and nonnegativity of the Leslie coefficients is exactly the condition $p_k \leq 0$. □

Set $c_0 = 1$ and $c_k = 0$ outside $0 \leq k \leq d$, and set $r_0 = 1$. Then

$$p_k = \sum_{j=0}^{q} r_j c_{k-j} = c_k + \sum_{j=1}^{q} c_{k-j} r_j. \tag{29.5}$$

Define the $(d+q) \times q$ Toeplitz matrix

$$(A_q)_{kj} = c_{k-j}, \qquad 1 \leq k \leq d+q, \quad 1 \leq j \leq q, \tag{29.6}$$

and the vector

$$h_q = (-c_1, \ldots, -c_d, 0, \ldots, 0)^T \in \mathbb{R}^{d+q}. \tag{29.7}$$

Theorem 29.3. *For fixed q, a degree-q auxiliary completion exists if and only if the linear system*

$$\boxed{A_q r \leq h_q} \tag{29.8}$$

is feasible for $r = (r_1, \dots, r_q)^T$. *Consequently,*

$$\boxed{\nu(C) = \min\{q \geq 0 : A_q r \leq h_q \text{ is feasible}\}.} \tag{29.9}$$

Proof. Equation (29.5) gives $p_k \leq 0$ exactly as the kth inequality in (29.8). Definition 29.1 then gives (29.9). □

Proposition 29.4 (Dual infeasibility certificate)**.** *Degree q is impossible if and only if there exists a vector $y \in \mathbb{R}^{d+q}$ satisfying*

$$\boxed{y \geq 0, \qquad A_q^T y = 0, \qquad h_q^T y < 0.} \tag{29.10}$$

Such a y is a Farkas certificate that no choice of q auxiliary roots can repair the rigid core.

Proof. This is Farkas' alternative applied to the system $A_q r \leq h_q$. □

Proposition 29.5. *If degree q is feasible, every degree $q' \geq q$ is feasible. Hence the feasible auxiliary degrees form either the empty set or the tail*

$$\{\nu(C), \nu(C)+1, \nu(C)+2, \dots\}. \tag{29.11}$$

Proof. If R_q is feasible, then zR_q is monic of degree $q+1$ and

$$C(z) z R_q(z) = z\, C(z) R_q(z).$$

Its nonleading coefficients are the previous nonpositive coefficients shifted by one place, together with a final zero. Iteration proves the claim. □

29.1. **Auxiliary defect profile.** For a quantitative profile before exact feasibility, define

$$E_q(C) = \min_{r \in \mathbb{R}^q,\ e \geq 0} \{e : A_q r - h_q \leq e\mathbf{1}\}. \tag{29.12}$$

Equivalently, E_q is the minimum possible largest positive coefficient of the completed product. To compare different b values, the same definition may be applied after the sign-preserving scaling $z = \sqrt{b}\, w$, which replaces c_k by $c_k / b^{k/2}$.

Proposition 29.6. *The auxiliary defect profile satisfies*

$$E_q(C) \geq 0, \qquad E_{q+1}(C) \leq E_q(C), \qquad E_q(C) = 0 \iff q \geq \nu(C). \tag{29.13}$$

Thus $q \mapsto E_q(C)$ records how the rigid obstruction is removed as auxiliary degrees are introduced, while $\nu(C)$ is its first zero.

Proof. Nonnegativity is built into (29.12). The construction $R_q \mapsto zR_q$ used in Proposition 29.5 preserves the coefficient violation and proves monotonicity. Finally, $E_q = 0$ is exactly feasibility of (29.8). □

29.2. **Exact one-auxiliary test.** Because the rigid core has

$$c_d = -b^{K+1} < 0, \tag{29.14}$$

a degree-one multiplier $R_1(z) = z + t$ must satisfy $t \geq 0$. Its remaining conditions are

$$c_1 + t \leq 0, \qquad c_k + t c_{k-1} \leq 0, \quad 2 \leq k \leq d. \tag{29.15}$$

Hence $\nu(C) = 1$ exactly when the interval of t values determined by (29.15) and $t \geq 0$ is nonempty. Explicitly, every index with $c_{k-1} < 0$ supplies a lower bound

$$t \geq -\frac{c_k}{c_{k-1}}, \tag{29.16}$$

every index with $c_{k-1} > 0$ supplies an upper bound

$$t \leq -\frac{c_k}{c_{k-1}}, \tag{29.17}$$

and $c_{k-1} = 0$ requires $c_k \le 0$. The first coefficient supplies $t \le -c_1$.

Remark 29.7. The rigid margin Δ_K and the completion index ν measure different properties. The sign of Δ_K decides $q = 0$ feasibility, but its magnitude need not determine how many auxiliary degrees are necessary. The appropriate comparison is therefore not only $-\Delta_K$ versus ν, but the full scaled defect profile

$$E_0, E_1, \dots, E_\nu = 0.$$

For the eight $K = 7$ configurations in Experiment 28.1, this profile can be computed from the already estimated phase sets without any new frequency estimation.

Experiment 29.8. For each of the eight evaluable $K = 7$ phase sets, form the fixed core polynomial $C = P_{b,\Omega}$ once. Starting at $q = 0$, solve (29.8) in increasing degree. For every infeasible q, retain a dual vector satisfying (29.10); at the first feasible degree retain the multiplier coefficients, the completed Leslie coefficients, the minimum coefficient, the primal residual, and the spectral residual. Report

$$(b, \Delta_7, j_*, \nu, E_0, E_1, \dots, E_\nu)$$

for every case. This protocol certifies the minimal auxiliary repair degree rather than merely producing one high-dimensional feasible completion.

30. Static completion versus data-compatible repaired recurrence

The completion problem of Section 29 starts from a prescribed critical core $C = P_{b,\Omega}$ and asks whether multiplication by a monic real polynomial can make all nonleading coefficients nonpositive. This is an exact static inverse-spectral problem. For a moving scalar sequence one needs an additional condition: the completed polynomial must also be the recurrence polynomial of the data.

Fix $K \ge 1$, an auxiliary degree $q \ge 0$, and $b > 1$, and put

$$D = 2K + 1 + q. \tag{30.1}$$

For an order-D scalar recurrence define

$$h_t^{(D)} = (u_{t-1}, \dots, u_{t-D}) \in \mathbb{R}^D, \tag{30.2}$$

and for a window beginning at n set

$$\mathsf{A}_{n,D} = \begin{pmatrix} h_{n+D}^{(D)} \\ \vdots \\ h_{n+2D-1}^{(D)} \end{pmatrix}, \qquad \mathsf{g}_{n,D} = \begin{pmatrix} u_{n+D} \\ \vdots \\ u_{n+2D-1} \end{pmatrix}. \tag{30.3}$$

Definition 30.1 (Data recurrence polynomial)**.** If $\det \mathsf{A}_{n,D} \neq 0$, define

$$a_n^{(D)} = \mathsf{A}_{n,D}^{-1} \mathsf{g}_{n,D} \tag{30.4}$$

and

$$P_{n,D}(z) = z^D - \sum_{\ell=1}^{D} a_{n,\ell}^{(D)} z^{D-\ell}. \tag{30.5}$$

For $\theta = (\Omega, r) \in (0,\pi)^K \times \mathbb{R}^q$, write

$$R_q(z; r) = z^q + r_1 z^{q-1} + \cdots + r_q \tag{30.6}$$

and define the coefficient map $\Phi_{b,K,q}$ by

$$C_{b,\Omega}(z) R_q(z; r) = z^D - \sum_{\ell=1}^{D} \Phi_{b,K,q}(\theta)_\ell z^{D-\ell}. \tag{30.7}$$

Let

$$\mathcal{M}^{\mathrm{rep}}_{b,K,q} = \operatorname{im} \Phi_{b,K,q}, \qquad \mathcal{M}^{\mathrm{rep},+}_{b,K,q} = \mathcal{M}^{\mathrm{rep}}_{b,K,q} \cap \mathbb{R}^D_+. \tag{30.8}$$

Theorem 30.2 (Exact recurrence-first membership test)**.** *Assume* $\det \mathsf{A}_{n,D} \neq 0$*. The data window has an exact fixed-*(b, K, q) *repaired Leslie realization if and only if*

$$a_n^{(D)} \in \mathcal{M}^{\mathrm{rep},+}_{b,K,q}. \tag{30.9}$$

Equivalently, the uniquely determined recurrence polynomial $P_{n,D}$ *factors as*

$$P_{n,D}(z) = (z-b) \prod_{k=1}^{K} \left(z^2 - 2\sqrt{b} \cos \omega_k \, z + b \right) R_q(z; r) \tag{30.10}$$

for some real Ω, r*, and its Leslie coefficients are nonnegative.*

Proof. Nonsingularity gives a unique order-D coefficient vector satisfying the D recurrence equations. A repaired fixed-(b, K, q) Leslie representation of that same window must therefore have exactly that coefficient vector. By definition of $\Phi_{b,K,q}$ this is equivalent to the factorization (30.10); Leslie nonnegativity is exactly $a_n^{(D)} \geq 0$. □

Remark 30.3 (Frequency-free root form)*.* For exact data one may first compute $P_{n,D}$ and then test whether it contains a real root $b > 1$ and K distinct conjugate pairs with squared modulus b. The remaining quotient is the auxiliary factor. If several critical subsets are available, their choice is a branch choice and must be fixed from past information or tracked continuously; future test values may not be used to select among them.

Warning 30.4 (Static LP completion is not a data fit)*.* The Toeplitz LP (29.8) answers whether a prescribed core C can be embedded in some larger positive Leslie spectrum. Unless the resulting product polynomial also equals $P_{n,D}$, it does not establish that the scalar data window follows that completed recurrence. This distinction is essential for every moving-window inference.

31. Geometry and codimension of the repaired family

Proposition 31.1 (Auxiliary degree does not remove the critical codimension)**.** *At a point* θ *where the Jacobian* $D\Phi_{b,K,q}(\theta)$ *has full column rank* $K + q$*, the fixed-b repaired family is locally a smooth immersed manifold of dimension*

$$K + q \tag{31.1}$$

inside $\mathbb{R}^D$*, and hence has codimension*

$$D - (K + q) = K + 1. \tag{31.2}$$

The codimension is independent of the auxiliary degree q*.*

Proof. The parameter domain has $K + q$ real coordinates. Full column rank gives local image dimension $K + q$ by the constant-rank theorem. Substituting $D = 2K + 1 + q$ gives $D - (K + q) = K + 1$. □

Remark 31.2*.* Each auxiliary coefficient adds one parameter and simultaneously raises the ambient recurrence degree by one. Therefore increasing q does not make a generic scalar window automatically compatible with a critical Leslie core. Auxiliary completion repairs a positivity obstruction; it does not erase the $K + 1$ fixed-b spectral constraints.

Warning 31.3 (Fixed degree is part of the state model)*.* The repaired mobility theorem below assumes the same q on adjacent windows. Allowing q to change alters the recurrence dimension and can create rank-deficient or nonminimal representations. A dynamic protocol must therefore fix q before test, or specify a separate causal dimension-changing theory. Silent per-window reselection of q is an additional information channel, not part of the theorem.

32. Version 9 sharpening of the repaired Leslie geometry

The full-rank statements of Section 31 were deliberately conditional. For the fixed-b critical family those conditions can be characterized explicitly. Write $r = \sqrt{b}$ and

$$Q_j(z) = z^2 - 2r\cos\omega_j\, z + b, \qquad C_\Omega(z) = (z-b)\prod_{j=1}^{K} Q_j(z).$$

For interior distinct phases, define

$$U_j(z) := \frac{\partial C_\Omega}{\partial \omega_j} = \kappa_j z\,\frac{C_\Omega(z)}{Q_j(z)}, \qquad \kappa_j = 2r\sin\omega_j \neq 0. \tag{32.1}$$

Theorem 32.1 (Exact tangent space of the critical core)**.** *If $0 < \omega_1 < \cdots < \omega_K < \pi$, then*

$$T_C\mathcal{C}_{b,K} = \operatorname{span}_{\mathbb{R}}\{U_1, \ldots, U_K\},$$

and the U_j are linearly independent. Hence the fixed-b critical core has rank exactly K throughout the interior simple-phase locus.

Proof. Differentiate each quadratic factor to obtain (32.1). If $\sum_j \alpha_j U_j = 0$, reduce modulo Q_ℓ. Every term except U_ℓ vanishes, while zC/Q_ℓ is coprime to Q_ℓ under the stated hypotheses. Thus $\alpha_\ell = 0$ for every ℓ. □

Let $R(z) = z^q + r_1 z^{q-1} + \cdots + r_q$ and let $\Phi(C,R) = CR$. Tangent vectors to the monic degree-q factor are identified with $\mathbb{R}[z]_{<q}$, so

$$D\Phi_{(C,R)}(U,V) = RU + CV. \tag{32.2}$$

Define

$$I(C,R) = \{j : Q_j \mid R\}, \qquad h = |I(C,R)|.$$

Theorem 32.2 (Exact kernel of the repaired multiplication differential)**.** *On the interior simple-phase locus,*

$$\ker D\Phi_{(C,R)} = \operatorname{span}\left\{\left(U_j, -\kappa_j z\frac{R}{Q_j}\right) : j \in I(C,R)\right\}. \tag{32.3}$$

Consequently,

$$\boxed{\dim\ker D\Phi = h, \qquad \operatorname{rank} D\Phi = K + q - h.} \tag{32.4}$$

In particular, sharing the Perron factor $(z-b)$ with R does not by itself lower the rank when b is fixed.

Proof. Write $C = GC_0$, $R = GR_0$ with $\gcd(C_0, R_0) = 1$. From $RU + CV = 0$ one obtains $R_0U + C_0V = 0$, hence $C_0 \mid U$ and $R_0 \mid V$. Expanding $U = \sum_j \alpha_j U_j$ and reducing modulo any $Q_\ell \nmid R$ forces $\alpha_\ell = 0$. Conversely, if $Q_j \mid R$, then $-RU_j/C = -\kappa_j z R/Q_j \in \mathbb{R}[z]_{<q}$, producing the displayed kernel vectors. Their independence follows from Theorem 32.1. □

32.1. **Critical resultant and intrinsic normal space.**

Define the critical resultant

$$\Delta_{\mathrm{crit}}(C,R) = \prod_{j=1}^{K} \operatorname{Res}(Q_j, R). \tag{32.5}$$

Then, away from phase collisions and endpoints,

$$D\Phi \text{ has full column rank } \iff \Delta_{\mathrm{crit}}(C,R) \neq 0. \tag{32.6}$$

The ordinary resultant $\operatorname{Res}(C,R)$ contains the additional factor $\operatorname{Res}(z-b,R)$; its vanishing therefore includes the component $R(b) = 0$, which is not automatically singular for the fixed-b restricted map. Three loci must be kept distinct: core discriminant, critical resultant, and repeated Perron factor.

Let $\mathcal{P}_{<D}$ denote the real polynomials of degree $< D$. The linearized normal space is

$$N_{(C,R)} = \frac{\mathcal{P}_{<D}}{R\,T_C\mathcal{C}_{b,K} + C\,\mathbb{R}[z]_{<q}}. \tag{32.7}$$

Corollary 32.3 (Exact regular codimension and singular linearized codimension)**.** *With $h = |I(C,R)|$,*

$$\boxed{\dim N_{(C,R)} = K + 1 + h.} \tag{32.8}$$

Thus the regular codimension $K+1$ of Proposition 31.1 is not merely a parameter count: it is the dimension of the cokernel of the coefficient deformation map. At a singular point $h > 0$, $K+1+h$ is the codimension of the linearized image*; higher-order germ dimension need not equal this number.*

32.2. **Global branches and semialgebraic strata.** For a polynomial P divisible by $z - b$, let $\mathcal{E}_b(P)$ be the set of distinct critical quadratic factors Q_ω of P with $0 < \omega < \pi$.

Proposition 32.4 (Number of simple critical factorizations)**.** *If P contains N distinct critical quadratic factors and all are simple, then the factorizations $P = C_\Omega R$ with exactly K ordered critical factors in C_Ω are in bijection with the K-subsets of $\mathcal{E}_b(P)$. Hence*

$$\#\Phi^{-1}(P) = \binom{N}{K}.$$

If exactly K simple critical factors occur, the critical branch is globally unique. If a selected factor occurs twice, the same Q_j remains in R and the representation lies on the singular locus of Theorem 32.2.

Using $x_j = 2\sqrt{b}\cos\omega_j$, the ordered phase domain and the Leslie coefficient inequalities are semialgebraic. Standard semialgebraic stratification therefore gives finitely many analytic strata and finitely many connected components. This does *not* imply that the global positive repaired region is connected; convexity of the auxiliary fiber for a fixed core is insufficient to prove connectedness of the admissible base of cores.

32.3. **Irreducibility, primitivity, and recurrence minimality.** For a nonnegative Leslie companion $L(a)$ of order D,

$$L(a) = \begin{pmatrix} a_1 & \cdots & a_D \\ 1 & 0 & \cdots \\ & \ddots & \ddots \end{pmatrix}, \qquad a_j \geq 0,$$

the graph structure gives the following exact classification.

Proposition 32.5 (Leslie irreducibility and period)**.** *The matrix $L(a)$ is irreducible if and only if $a_D > 0$. On the irreducible stratum its period is*

$$\gcd\{j : a_j > 0\},$$

and it is primitive exactly when this gcd equals 1. For $P = CR$ with $q \geq 1$,

$$\boxed{a_D = b^{K+1} r_q,} \tag{32.9}$$

so a nonnegative repaired Leslie matrix is irreducible exactly when $r_q > 0$. For $q = 0$ one has $a_D = b^{K+1} > 0$.

Proposition 32.6 (Nonsingular recurrence matrix implies finite-window minimality)**.** *If the order-D recurrence matrix $\mathsf{A}_{n,D}$ is nonsingular, then no recurrence of order $d < D$ can hold throughout the rows required to construct $\mathsf{A}_{n,D}$. Hence the recovered order D is minimal on that finite recurrence window.*

Proof. If an order-d relation held, the first column of $\mathsf{A}_{n,D}$ would be a linear combination of columns $2,\dots,d+1$, contradicting nonsingularity. □

Corollary 32.7 (Zero-padding is static completion, not a minimal dynamic repair)**.** *Replacing a feasible multiplier R by z^hR preserves static nonnegative completion when the coefficient inequalities allow it, but the polynomial z^hP merely appends h zero lag coefficients. It therefore cannot create a new recurrence-first minimal realization of order $D+h$ for data already governed by the order-D recurrence.*

33. Exact repaired overlap and nonlinear scalarization

Consider two order-D recurrence windows beginning at n and $n+s$, with $1 \le s \le D$. They share exactly $D-s$ recurrence equations. Define

$$O_{n,s}^{(D)} = \begin{pmatrix} h_{n+s+D}^{(D)} \\ \vdots \\ h_{n+2D-1}^{(D)} \end{pmatrix} \in \mathbb{R}^{(D-s)\times D}. \tag{33.1}$$

Let the old coefficient vector be $a_n = \Phi_{b,K,q}(\theta_n) \in \mathcal{M}_{b,K,q}^{\mathrm{rep},+}$. Every candidate next coefficient vector on the same repaired branch that satisfies all common equations must obey

$$F_{n,s}(\theta) := O_{n,s}^{(D)}\left(\Phi_{b,K,q}(\theta) - a_n\right) = 0. \tag{33.2}$$

Theorem 33.1 (Local repaired mobility dimension)**.** *Assume $D\Phi_{b,K,q}(\theta_n)$ has rank $K+q$ and that $DF_{n,s}(\theta_n)$ has maximal possible rank $\min\{K+q, D-s\}$. Then the local solution set of* (33.2) *through θ_n has dimension*

$$\boxed{\max\{0, s-K-1\}.} \tag{33.3}$$

Proof. The parameter space has dimension $K+q$. The constant-rank theorem gives local level-set dimension $K+q-\min\{K+q, D-s\}$. Since $D = 2K+1+q$, this equals $\max\{0, s-K-1\}$. □

Corollary 33.2 (Repaired critical stride)**.** *Under the hypotheses of Theorem 33.1, the first generic one-dimensional repaired motion occurs at*

$$\boxed{s_* = K+2,} \tag{33.4}$$

independently of q.

Remark 33.3. For $q=0$ and fixed b, the reciprocal–Perron affine hull makes the critical motion the exact line $a+\tau v$ of Section 12. For $q>0$, the repaired coefficient family is generally nonlinear. The one-dimensional object supplied by Corollary 33.2 is therefore a local curve, not in general an affine line. Replacing that curve by its tangent is only a first-order approximation.

At the repaired critical stride define the first genuinely new index

$$t_0 = n+2D. \tag{33.5}$$

All entries of $h_{t_0}^{(D)}$ are known before u_{t_0} is revealed.

Theorem 33.4 (One observed scalar selects the local repaired continuation)**.** *Assume the hypotheses of Corollary 33.2, and let Γ_n be the regular one-dimensional local solution branch of* (33.2) *through θ_n. Define*

$$\chi_n(\theta) = h_{t_0}^{(D)}\Phi_{b,K,q}(\theta), \qquad \theta \in \Gamma_n. \tag{33.6}$$

If the differential of χ_n does not vanish on the tangent line $T_{\theta_n}\Gamma_n$, then χ_n is locally one-to-one on Γ_n. Consequently, every sufficiently near next realization on the same branch is uniquely selected by the single observed value u_{t_0}. A canonical scalar innovation is

$$\boxed{\xi_n = u_{t_0} - h_{t_0}^{(D)} a_n.} \tag{33.7}$$

Locally there is a unique map $\theta^+ = \Theta_n(\xi_n)$ and hence

$$a^+ = \Phi_{b,K,q}(\Theta_n(\xi_n))\,. \tag{33.8}$$

Proof. The regular common-equation set is a one-dimensional manifold by Corollary 33.2. A nonzero derivative of the scalar map χ_n along its tangent makes χ_n a local coordinate by the one-dimensional inverse function theorem. The observed value u_{t_0} therefore selects one nearby point on the branch. Equation (33.7) is a translated scalar coordinate. □

Corollary 33.5 (Repaired $K+1$-value causal forecast)**.** *Under Theorem 33.4, assume the true next window lies on the selected local branch and uses the coefficient vector a^+ of* (33.8)*. After revealing only u_{t_0}, the remaining*

$$s_* - 1 = K + 1 \tag{33.9}$$

new values of the next window are determined recursively by the order-D recurrence with coefficient vector a^+.

Warning 33.6 (Local versus finite-step continuation). Theorem 33.4 is local. A finite transition must additionally certify that the continuation stays on one regular connected branch, remains nonnegative, preserves the chosen critical factor, and does not cross a point where the overlap rank or the observation derivative degenerates. At such an event the correct status is `NOT_APPLICABLE` or `NUMERICALLY_AMBIGUOUS`, not branch reselection from future data.

33.1. **Version 9 observability certificates and global branch uniqueness.** At the repaired critical stride let $p = K + q$ and write

$$J = DF_{n,s_*}(\theta) = O_{n,s_*}^{(D)} D\Phi(\theta), \qquad g = h_{t_0}^{(D)} D\Phi(\theta).$$

Assume $\operatorname{rank} J = p - 1$, so the common-equation fiber is locally a regular curve.

Theorem 33.7 (Equivalent transversality certificates)**.** *The following are equivalent:*

$$d\chi_n|_{\ker J} \neq 0, \qquad \ker J \cap \ker g = \{0\}, \qquad \boxed{\operatorname{rank}\begin{pmatrix} J \\ g \end{pmatrix} = p.}$$

Thus the phrase "the first-new observation is transverse" has an exact augmented-rank certificate.

Theorem 33.8 (Global injectivity on one regular connected branch)**.** *Let Γ be a connected regular one-dimensional component of the common-equation set. If $d\chi_n$ never vanishes on $T\Gamma$, then*

$$\chi_n : \Gamma \longrightarrow \mathbb{R}$$

is globally injective.

Proof. A connected one-dimensional manifold is locally an interval; on every interval coordinate the continuous nonzero derivative has constant sign, so χ_n is strictly monotone. A circular component cannot support a real-valued smooth function with nowhere-vanishing derivative because an extremum must occur. □

Remark 33.9 (Failure classification). Loss of scalar observability can occur through distinct mechanisms: an observation fold, $\operatorname{rank} J < p - 1$, a critical-factor transfer $\Delta_{\text{crit}} = 0$, a core collision $\operatorname{Disc}(C) = 0$, or contact with a Leslie face $a_\ell = 0$. These events must be reported separately because they have different geometric meanings.

34. Causal state and hidden-information audit

Proposition 34.1 (Rigid direction is past-determined)**.** *In the rigid fixed-b case at $s_* = K + 2$, suppose $\ker Z_{n,s_*}$ is one-dimensional and $h_{t_0} v \neq 0$. The normalization*

$$\|v_n\|_2 = 1, \qquad h_{t_0} v_n > 0 \tag{34.1}$$

makes v_n a deterministic function of data available before u_{t_0}. Its variation from one window to another is therefore not, by itself, an additional external information channel.

Proof. Both O_{n,s_*} and R_b are determined before u_{t_0}, hence so is the one-dimensional nullspace. Unit norm removes scale and the sign condition removes the remaining sign ambiguity. □

Proposition 34.2 (Auxiliary coefficients are not independent innovations on a regular repaired branch)**.** *Fix K, q, b and a regular local branch as in Theorem 33.4. Then all components of $\theta^+ = (\Omega^+, r^+)$ are locally functions of the single scalar ξ_n. In particular, the q auxiliary coefficients r^+ do not supply q independent new parameters on that transition.*

Proof. Theorem 33.4 gives the local map $\theta^+ = \Theta_n(\xi_n)$. Projection onto the r coordinates gives $r^+ = \pi_r \Theta_n(\xi_n)$. □

Warning 34.3 (When hidden channels re-enter)*.* Proposition 34.2 fails as an information statement if any of the following is selected anew using unseen data:

(1) the auxiliary degree q;
(2) the Perron value b when no causal rule for b has been fixed;
(3) a disconnected or singular factorization branch;
(4) an independently optimized auxiliary multiplier in each window;
(5) a window-specific normalization, forcing term, observation operator, or state reset.

Every such choice must either be part of the past state with a deterministic update rule or be counted as an additional innovation.

Remark 34.4 (The minimum-norm LP selector)*.* A minimum-norm solution of the static LP is a legitimate deterministic selector for one fixed core, but applying it independently to every new core does not prove that the resulting sequence is the continuation curve of Section 33. Static canonicality and dynamic causality are different requirements.

35. Information accounting

The phrase "one innovation" is a statement about local degrees of freedom, not about a finite total amount of information in an infinite trajectory.

Proposition 35.1 (Conditional block determinacy)**.** *Let X_n denote the complete past-determined state of a regular fixed-(K, q, b) repaired transition, including the selected local branch. Under Theorem 33.4, there is a deterministic map G_n such that*

$$(u_{t_0}, u_{t_0+1}, \dots, u_{t_0+K+1}) = G_n(X_n, \xi_n). \tag{35.1}$$

Moreover ξ_n is itself determined by X_n and u_{t_0} through (33.7)*. Hence, after X_n and the first new scalar are known, the remaining $K + 1$ values contain no further exact degrees of freedom within the model.*

Remark 35.2 (Entropy formulation)*.* If a discrete or explicitly quantized stochastic model is imposed, exact block determinacy gives

$$H(U_{t_0:t_0+K+1} \mid X_n) = H(U_{t_0} \mid X_n)\,, \tag{35.2}$$

because the remaining block is a deterministic function of (X_n, U_{t_0}). For continuous variables, differential entropy and finite-resolution coding require additional care, so no bit count is asserted without a declared quantization or probability model.

Remark 35.3 (Unbounded temporal information remains possible). The sequence (ξ_n) can itself contain new information at every transition. Thus finite state dimension does not imply finite information for the full infinite trajectory. A law for (ξ_n) is needed for a zero-latency generative model, but it is *not* needed to test the weaker and already nontrivial claim that one revealed scalar makes the next $K+1$ values redundant.

36. Why the previous approximate scalarization is not a coherent relaxation

The exact rigid theorem assumes that the old coefficient vector already satisfies the common recurrence equations and lies in the fixed-b reciprocal–Perron affine hull. Projecting an inexact least-squares coefficient vector onto the affine hull and then moving in the exact overlap kernel does not repair a pre-existing common-equation residual.

Proposition 36.1 (Overlap residual is invariant along the rigid null direction)**.** *Let O be the common-equation matrix, let y be the corresponding right-hand side, and let $v \in \ker O$. Then for every coefficient vector a and every scalar τ,*

$$O(a+\tau v) - y = Oa - y. \tag{36.1}$$

Therefore a nonzero common-equation residual cannot be corrected by choosing τ.

Proof. Subtracting gives $\big(O(a+\tau v)-y\big) - (Oa-y) = \tau Ov = 0$. □

Corollary 36.2. *A causal forecast based on a projected or constrained least-squares base a_C with $Oa_C \neq y$ and an exact null direction $v \in \ker O$ is not a numerical relaxation of Theorem 12.2. Its forecast error may diagnose that chosen projection procedure, but it does not by itself falsify or verify the exact scalarization theorem.*

A coherent approximate model must instead be posed directly on the nonlinear admissible family, for example by minimizing

$$\|\mathsf{A}_{n,D}\Phi_{b,K,q}(\theta) - \mathsf{g}_{n,D}\|_W \tag{36.2}$$

using past data only, and by propagating the full set of overlap-compatible parameters satisfying a predeclared residual tolerance. If that set is not effectively one-dimensional or the prediction interval is broad, the output is ambiguous rather than a forced scalar forecast.

Remark 36.3 (Conditioning). Near a regular exact point, the smallest nonzero singular value of the overlap Jacobian controls first-order sensitivity of the recovered branch to perturbations. Consequently every approximate implementation should report the singular spectrum and stability under row/column scaling and perturbation; a binary numerical rank alone is insufficient.

37. Audit of the 5 September 2026 Mertens computation

This section records the output of `leslie_mobius_information_audit_v1_1` and reinterprets it using the corrected distinctions above. The purpose is to freeze a baseline, not to promote approximate diagnostics to theorems.

Experiment 37.1 (Frozen v1.1 baseline)**.** The observable used 161 values on a geometric grid with sampling base $B_{\mathrm{obs}} = 1.1$, split into 96 training and 65 test points. Training-only model selection over $K \in \{3,4,5,6,7\}$ and 41 Perron candidates selected

$$K = 4, \qquad m = 9, \qquad s_* = 6, \qquad b = 1.02. \tag{37.1}$$

The selected b was exactly the lower endpoint of the searched grid.

For the strict rigid test, no training or test window satisfied all required conditions. In particular the exact fixed-b reciprocal test and nonnegativity test had zero accepted windows in both partitions. Because the projected base vectors were not nonnegative, the theorem-level interval I_{adm}

containing $\tau = 0$ was not available; the reported zero interval coverage is therefore only a diagnostic of the projected procedure. The approximate projected pipeline nevertheless produced

$$\mathrm{NRMSE}_{\mathrm{Leslie,test}} = 6.3668, \tag{37.2}$$

$$\mathrm{NRMSE}_{\mathrm{persistence,test}} = 1.4657, \tag{37.3}$$

$$\mathrm{NRMSE}_{\mathrm{AR,test}} = 0.67843. \tag{37.4}$$

Leslie beat AR in none of the ten evaluated test transitions.

The numerical identity reconstructing the first new observation from the fitted scalar innovation had maximum relative discrepancy 2.689×10^{-17}, which confirms only that the scalar formula recodes the first newly observed value. The ten test innovations gave the exploratory entropy ratio

$$\widehat{H}(\tau_n \mid \tau_{n-1})/\widehat{H}(\tau_n) = 0.3142 \tag{37.5}$$

and an AR prediction ratio for τ of 1.8136 relative to the median predictor. The sample size is too small for the entropy estimate to support a law for τ.

The median angle between successive normalized rigid null directions was 64.8726°. By Proposition 34.1, this large variation is not itself evidence of an extra information channel.

For the ten projected critical cores used by the approximate pipeline, no static auxiliary completion with $q \le 4$ was found, and all 40 attempted positive degrees $q = 1, \dots, 4$ had numerical Farkas certificates. This establishes a lower-degree obstruction for those *projected cores*; because those cores were not exact recurrence polynomials of the data, it does not establish $\nu > 4$ for a data-compatible repaired Mertens model.

Finally, an optional exact symbolic calculation on the unnormalized integer values $M(x)$ examined two nonsingular order-9 windows. In both, the gcd of the $K + 1$ polynomial fixed-Perron constraints was constant. For those two raw windows and $K = 4$, this excludes any b satisfying all rigid reciprocal–Perron identities.

Remark 37.2 (Null-model comparison). With the same frozen approximate model, the median test forecast errors across 100 null realizations were approximately 2.354 for support-preserving random signs and 2.464 for permuted Möbius values, both substantially below the Möbius value 6.3668. The admissible-interval fraction was zero for Möbius and both null classes. The entropy-ratio and null-direction-angle diagnostics did not produce a favorable Möbius separation. These observations are negative for that approximate protocol, not for the exact repaired theory developed in Sections 30–35.

Remark 37.3 (Operational consequence). The selected $b = 1.02$ cannot be interpreted as a discovered arithmetic constant because it lies on the search boundary. Extending the numerical grid toward 1 before repairing the model would amount to further adaptation after a failed out-of-sample test. The baseline (37.1)–(37.4) should instead remain frozen for later regression testing.

38. Revised causal protocol for repaired windows

Experiment 38.1. Before any test window is examined:

1. fix K, a single auxiliary degree q, and either a fixed b or a deterministic past-only rule for b covered by a separate theorem;
2. fix the scalar observable, window convention, stride $s_* = K + 2$, numerical tolerances, branch-continuation rule, and forecast metric;
3. validate the exact algebra and continuation code on synthetic trajectories for which the true (Ω, r) path is known;
4. on each real window, construct the order-D recurrence from the data first and require the membership test of Theorem 30.2;
5. reject or mark ambiguous every singular, nonpositive, branch-ambiguous, or ill-conditioned state;

(6) construct the common-equation branch from past values only;
(7) reveal only u_{t_0}, compute the canonical innovation ξ_n, and solve for the next point on the same branch without using later test values;
(8) predict the remaining $K+1$ values and compare against predeclared baselines and nulls.

A successful repaired experiment must therefore establish both *membership* and *transport*. Static completion of an externally estimated phase set is not a substitute for either step.

39. Version 4 predictive audit: closure of the arithmetic-free route

The recurrence-first and nonlinear-continuation corrections of Version 7 were subsequently implemented in a stricter predictive instrument. The synthetic gate was first required to pass before the Mertens data were inspected. Three exact synthetic oracles passed:

$$(K,q,D) = (1,0,3), \qquad (2,1,6), \qquad (3,2,9).$$

In the $q=2$ case the five-dimensional parameter $(\omega_1,\omega_2,\omega_3,r_1,r_2)$ was recovered from the overlap plus one new scalar with relative state error approximately 1.45×10^{-8} and subsequent forecast error approximately 1.37×10^{-15}. Thus the repaired nonlinear scalarization mechanism itself survived a nontrivial synthetic test.

The corresponding first-sample conditioning diagnostics were

(K,q,D)	observability margin	rcond	forecast relative error
$(1,0,3)$	9.958×10^{-1}	1.000	0
$(2,1,6)$	5.613×10^{-3}	1.132×10^{-5}	2.599×10^{-12}
$(3,2,9)$	9.145×10^{-7}	7.813×10^{-9}	1.369×10^{-15}

so the exact mechanism survives while becoming increasingly ill-conditioned in these particular synthetic examples. This conditioning trend is a numerical warning, not a failure of the local dimension count.

Experiment 39.1 (Frozen Version 4 Mertens audit)**.** The real-data protocol kept

$$K=4, \qquad b=1.02$$

frozen from the previous train-only selection and tested auxiliary degrees $q=0,1,\dots,6$. The exact recurrence-first membership count was

$$\boxed{0 \text{ strict windows}}$$

in both train and test for every tested q. The detailed frozen recurrence-first scan was

q	D	N_{train}	$N_{\text{strict,train}}$	$\widetilde{s}_{\text{train}}$	N_{test}	$\widetilde{s}_{\text{test}}$
0	9	13	0	0.8761	11	1.084
1	10	13	0	1.106	11	0.8054
2	11	13	0	0.8857	11	0.9118
3	12	12	0	0.9261	11	1.211
4	13	12	0	0.9057	11	0.8922
5	14	12	0	1.292	11	1.102
6	15	11	0	2.500	11	1.197

where $\widetilde{s}$ denotes the median membership score. These values are of order one rather than numerical noise. In the corrected positive approximate nonlinear-manifold layer, the reported test errors were

$$\text{NRMSE}_{\text{Leslie,assisted}} \approx 0.909, \qquad \text{NRMSE}_{\text{AR,direct}} \approx 0.646,$$

so the Leslie construction was approximately 1.374 times the direct AR error.

The primary predictive comparisons did not separate Mertens from the stronger controls: support-preserving random-sign comparisons gave empirical p-values approximately 0.235 and 0.196 on the

reported primary scores, and the phase-randomized surrogate comparison gave approximately 0.192. Only the simpler Gaussian white null produced a nominally small value, approximately 0.0196.

Remark 39.2 (Interpretation of the negative result). Experiment 39.1 does not prove that no Leslie representation of Mertens can exist. It closes the specific arithmetic-free route in which the frozen critical geometry, fixed Perron value, and auxiliary degrees through $q = 6$ were asked to discover a predictive law from the scalar Mertens observable alone. The combination

$$\text{zero exact membership} + \text{worse forecast than direct AR}$$
$$+\text{no separation from phase surrogates}$$

provides no empirical reason to continue increasing geometric flexibility without supplying arithmetic structure.

Remark 39.3 (Why the small first-value residual is not predictive evidence). The Version 4 approximate continuation reported a median first-value residual about 0.1336, compared with a common-equation residual about 0.9567. This factor-of-seven difference is not a sevenfold predictive gain: the observed first value u_{t_0} enters the nonlinear objective used to select the assisted next state. The quantity is therefore a conditional fitting residual. The zero-latency innovation predictor is the relevant non-tautological test. Its reported RMSE was about 0.1634. The corresponding empirical values were approximately 0.0196 against support-preserving random signs, 0.0392 against the Gaussian null, and 0.3077 against the phase-randomized surrogate. The failure against the phase surrogate removes the arithmetic interpretation of the small RMSE. No further rescue test of the 0.1336 residual is required for the present branch.

40. Finite-range cancellation diagnostic only

This computation is retained because it was part of the numerical history, but it is *not* an asymptotic baseline and it plays no role in the active arithmetic-driven argument. For

$$1000 \le x \le 455\,052\,511$$

the exact finite-range maximum observed in that computation was

$$\max_x \frac{|M(x)|}{\sqrt{x}} = 0.472203, \tag{40.1}$$

attained at $x = 2803$. At the final endpoint,

$$M(455\,052\,511) = -5414,$$

and the squarefree-support density was 0.607927.

The decade maxima recorded in the same finite run were

$10^3 \le x < 10^4$	0.472203
$10^4 \le x < 10^5$	0.462977
$10^5 \le x < 10^6$	0.437776
$10^6 \le x < 10^7$	0.418245
$10^7 \le x < 10^8$	0.462729
$10^8 \le x < 10^9$	0.464162

within the available range.

On the same squarefree support, 25 global random-sign realizations had mean maximum constant 1.8972, standard deviation 0.3384, minimum 1.1371, and maximum 2.7911. None fell below (40.1); the observed ratio of the null mean to the finite-range Mertens maximum was about 4.018.

Remark 40.1 (Why this is not a baseline for the size of Mertens). Known results already force the normalized Mertens function outside the scale seen in (40.1): the currently recorded explicit limsup/liminf bounds include

$$\limsup_{x\to\infty} \frac{M(x)}{\sqrt{x}} > 1.826054, \qquad \liminf_{x\to\infty} \frac{M(x)}{\sqrt{x}} < -1.837625.$$

Thus the value 0.472203 is only a local feature of the computed interval and cannot be used as an envelope, a stability constant, or evidence for square-root boundedness. In particular, there are arbitrarily large arguments along which the normalized function exceeds fixed positive and negative constants far larger in magnitude than the finite-range value above.

It is important not to strengthen this statement beyond what is known unconditionally: unboundedness of $|M(x)|/\sqrt{x}$ is widely expected, but is not presently an unconditional theorem.

Warning 40.2. The finite-range random-sign comparison is retained only as a diagnostic of that particular data range. It is too weak to support an asymptotic claim and too weak to distinguish genuinely arithmetic structure from spectral or cancellation structure already present in the observable. Accordingly, it has no evidential weight in the Version 8 arithmetic-driven Leslie tower.

41. Periodicity, lifted states, and what moving matrices do not guarantee

A moving sequence L_n has no mandatory spectral periodicity, but time dependence alone does not exclude cycles. Three different notions must be separated.

Definition 41.1 (Three periodicities)**.** Let X_n be the complete dynamical state, let $L_n = \mathcal{L}(X_n)$, and let u_n be the scalar observable.

(1) Matrix periodicity of period p: $L_{n+p} = L_n$.
(2) Lifted-state periodicity of period p: $X_{n+p} = X_n$.
(3) Observable periodicity of period p: $u_{n+p} = u_n$.

None of these implications is automatic in both directions.

For spectral tracking, the state must contain enough information to identify the continued factorization branch, not only the coefficient vector or the cosines of the critical phases. We retain the historical notation

$$X_n = \left(a_n, \widetilde{\Omega}_n, r_n, \mathfrak{b}_n\right), \tag{41.1}$$

but $\widetilde{\Omega}_n$ is to be read as a branch-aware continuation coordinate, not as an artificial 2π unwrapping of phases that are intrinsically ordered in $(0,\pi)$. The branch label $\mathfrak{b}_n$ records which critical factors are being continued. Equality of instantaneous coefficient vectors can therefore represent a projected return without a true return of the complete continuation state.

Proposition 41.2 (Return equation)**.** *Suppose a causal transition has the form*

$$X_{n+1} = F(X_n, \xi_n).$$

A genuine period-p return requires

$$F_{\xi_{n+p-1}} \circ \cdots \circ F_{\xi_n}(X_n) = X_n. \tag{41.2}$$

Thus periodicity is an additional fixed-point condition on the innovation word and the state; it is not a consequence of allowing L_n to vary.

Remark 41.3 (Periodic coefficients need not give a periodic observable). Even if $L_{n+p} = L_n$, the one-period monodromy

$$\Pi_p = L_{p-1} \cdots L_0$$

controls the evolution over complete periods. Unless the relevant monodromy action itself closes on the initial state, the generated scalar sequence need not be periodic. Hence matrix periodicity, lifted-state periodicity, and observable periodicity must not be conflated.

The arithmetic filtration introduced below supplies a second way to avoid a false finite-state periodicity argument: the state dimension itself grows as new divisor layers are admitted. This does not prove aperiodicity, but it removes any requirement that the entire arithmetic trajectory be generated by one fixed finite-dimensional spectrum.

42. Methodological firewall for the arithmetic continuation

The continuation below deliberately introduces arithmetic, but only through exact finite divisibility relations. The firewall is operational rather than ontological.

Definition 42.1 (Divisibility-only route)**.** For the active arithmetic route, the allowed primitives are finite divisor incidence, restriction/extension maps of finite divisor posets, integer matrix inversion when exact, and identities retaining every floor function exactly. The route does not invoke Dirichlet generating series, Mellin transforms, zeta functions, zero sets, or a replacement

$$\left\lfloor \frac{x}{d} \right\rfloor \mapsto \frac{x}{d} + \text{error}.$$

Remark 42.2. The identity

$$\sum_{d|n} \mu(d) = \delta_{n,1}$$

and the associated incidence-algebra inversion are classical. Their use here is not claimed as new. The research question is whether this parameter-free arithmetic source can be represented by a coherent tower of the previously developed positive Leslie systems without independent refitting at each level.

43. The finite divisibility system

Fix $X \geq 1$ and define the $X \times X$ incidence matrix

$$(D_X)_{n,d} = \mathbf{1}_{d|n}, \qquad 1 \leq n, d \leq X. \tag{43.1}$$

It is lower triangular with unit diagonal.

Theorem 43.1 (Exact finite Möbius inversion)**.** *The matrix D_X is unimodular and*

$$(D_X^{-1})_{n,d} = \begin{cases} \mu(n/d), & d \mid n, \\ 0, & d \nmid n. \end{cases} \tag{43.2}$$

Consequently, with

$$\mu_X = (\mu(1), \ldots, \mu(X))^{\mathsf{T}},$$

one has

$$D_X \mu_X = e_1, \qquad \mu_X = D_X^{-1} e_1, \tag{43.3}$$

and

$$M(X) = \mathbf{1}^{\mathsf{T}} D_X^{-1} e_1. \tag{43.4}$$

Proof. For $d \mid n$, the (n, d) entry of the proposed product $D_X D_X^{-1}$ is

$$\sum_{d|m|n} \mu(n/m) = \sum_{r|n/d} \mu(r),$$

which equals 1 when $n = d$ and 0 otherwise. The triangular unit diagonal gives $\det D_X = 1$. □

Remark 43.2 (Forced input and output). In the divisibility realization there is no free observation vector or initial vector:

$$\boxed{h = \mathbf{1}, \qquad v = e_1.}$$

The contrast with the earlier Leslie fits is fundamental. Here both vectors are fixed before M is inspected.

Summing (43.3) over rows $n \le x$ yields the exact floor identity

$$\sum_{d \le x} \mu(d) \left\lfloor \frac{x}{d} \right\rfloor = 1. \tag{43.5}$$

Within the divisibility-only route, the floor in (43.5) is never relaxed.

44. A coherent filtration by divisor rank

Let $\Omega(n)$ denote the number of prime factors counted with multiplicity. Equivalently for the present purpose, $\Omega(n)$ is the number of cover steps in a longest saturated divisor chain from 1 to n. For fixed X, define

$$I_k(X) = \{n \le X : \Omega(n) \le k\}, \qquad S_{k+1}(X) = I_{k+1}(X) \setminus I_k(X). \tag{44.1}$$

Because $d \mid n$ implies $\Omega(d) \le \Omega(n)$, each $I_k(X)$ is an order ideal and

$$I_0(X) \subset I_1(X) \subset \cdots .$$

For fixed X the tower reaches all integers by $k = \lfloor \log_2 X \rfloor$, since 2^k minimizes an integer with $\Omega(n) = k$.

Warning 44.1 (Maximum level versus typical level). The exact maximum filtration height is $\lfloor \log_2 X \rfloor$. It must not be replaced by a heuristic "typical" order such as $\log \log X$ inside an exact induction argument. No probabilistic distribution theorem for Ω is used in the present route.

Let $D_k(X)$ be the restriction of D_X to rows and columns indexed by $I_k(X)$.

Theorem 44.2 (Exact block extension of the divisor filtration)**.** *After ordering $I_{k+1}(X)$ with the old ideal first and the new layer second,*

$$D_{k+1} = \begin{pmatrix} D_k & 0 \\ B_k & I \end{pmatrix}, \tag{44.2}$$

where B_k is the incidence block from old divisors to the new layer. Therefore

$$D_{k+1}^{-1} = \begin{pmatrix} D_k^{-1} & 0 \\ -B_k D_k^{-1} & I \end{pmatrix}. \tag{44.3}$$

If

$$\mu_k = D_k^{-1} e_1,$$

then

$$\boxed{\mu_{k+1} = \begin{pmatrix} \mu_k \\ -B_k \mu_k \end{pmatrix}.} \tag{44.4}$$

In particular,

$$\mu_{k+1}|_{I_k} = \mu_k. \tag{44.5}$$

Proof. The upper-right block is zero because a new-layer element cannot divide an old element. If $d, n \in S_{k+1}(X)$ and $d \mid n$, then $\Omega(d) = \Omega(n)$, hence $\Omega(n/d) = 0$ and $d = n$. The new-layer diagonal block is therefore the identity. The inverse follows by direct block multiplication, and (44.4) follows by applying the inverse to e_1, which belongs to the old block. □

Theorem 44.3 (Canonical continuation operator)**.** *Let $E_k : \mathbb{R}^{|I_k|} \to \mathbb{R}^{|I_{k+1}|}$ be the zero-extension $E_k x = (x, 0)$. Define*

$$U_k := D_{k+1}^{-1} E_k D_k. \tag{44.6}$$

Then

$$\boxed{U_k x = \begin{pmatrix} x \\ -B_k x \end{pmatrix}}, \qquad \boxed{D_{k+1} U_k = E_k D_k}, \qquad \boxed{\operatorname{Res}_k U_k = I}. \tag{44.7}$$

Thus the divisor filtration carries an exact non-autonomous cocycle of injective extension maps, with no free innovation once B_k is given.

Proof. Insert the block inverse (44.3) into (44.6). The three identities follow by direct multiplication. □

Corollary 44.4 (Exact location of new arithmetic information)**.** *No previously determined Möbius value is altered when the filtration grows. The entire new layer is*

$$\boxed{\Delta_{k+1}^{\mathrm{arith}} = -B_k \mu_k.} \tag{44.8}$$

Thus the sequence of incidence blocks

$$B_0, B_1, B_2, \dots$$

is a parameter-free source of new arithmetic constraints.

Corollary 44.5 (Explicit value of the new Möbius layer)**.** *For $n \in S_{k+1}(X)$,*

$$(-B_k \mu_k)(n) = \mu(n) = \begin{cases} (-1)^{k+1}, & n \text{ squarefree}, \\ 0, & p^2 \mid n \text{ for some prime } p. \end{cases} \tag{44.9}$$

Hence the full incidence block can have high algebraic complexity while its action on the particular Möbius state returns the squarefree support of the new layer with one global sign. Cancellation in M_k remains inter-layer rather than internal to this new vector.

Remark 44.6 (Directed compatibility in both k and X). The pairs (k, X) form a directed system under coordinatewise inclusion, and the finite order ideals $I_k(X)$ exhaust $(\mathbb{N}_{\geq 1}, |)$. Restriction between finite ideals preserves incidence convolution, so the compatible finite inverses recover the usual Möbius element in the projective limit. When k increases by one at fixed X, the new diagonal block is exactly I as above. When X increases at fixed k, the newly added block is only unitriangular in general because two newly admitted integers may divide one another. These two growth directions must not be conflated.

This resolves a conceptual issue left open by the arithmetic-free Leslie programme. An infinite trajectory need not hide its information inside arbitrary real innovations. The information may enter by exact finite extensions of a growing arithmetic state, with coherence enforced by (44.5).

45. Why the naive level envelope is not the right induction variable

Define the truncated level sum

$$M_k(x) = \sum_{\substack{n \leq x \\ \Omega(n) \leq k}} \mu(n). \tag{45.1}$$

Also let

$$Q_j(x) = \#\{n \leq x : \ n \text{ is squarefree}, \ \Omega(n) = j\}. \tag{45.2}$$

Then, purely from the definition of μ,

$$M_k(x) = \sum_{j=0}^{k} (-1)^j Q_j(x), \qquad M_k(x) - M_{k-1}(x) = (-1)^k Q_k(x). \tag{45.3}$$

Proposition 45.1 (No within-layer cancellation)**.** *Every nonzero Möbius value in the squarefree layer $\Omega(n) = k$ has the same sign $(-1)^k$. Consequently, cancellation in M_k occurs between different filtration layers, not within the new layer itself.*

This makes a direct induction on

$$C_k(X) = \max_{x \le X} \frac{|M_k(x)|}{\sqrt{x}}$$

a poor first invariant. A small C_{k+1} would have to result from cancellation between the old partial sum and an entire fixed-sign layer; it is not generated by a contractive norm on the new layer alone. Normalizing a single layer by its own density does not create cancellation.

A more faithful elementary object is the paired-layer increment

$$P_j(x) = Q_{2j}(x) - Q_{2j-1}(x), \tag{45.4}$$

for which

$$M_{2J}(x) = 1 + \sum_{j=1}^{J} P_j(x). \tag{45.5}$$

No bound is asserted here. The point is only that (45.4) places opposite Möbius signs inside the same object before any norm is applied.

46. An exact prime-injection route inside the same firewall

The Ω filtration is the primary dimension-growing route. A second exact construction, useful as a controlled auxiliary experiment, introduces prime constraints one at a time without using a Dirichlet series.

For a prime p, define the finite divisor-shift operator

$$(T_p f)(n) = \begin{cases} f(n/p), & p \mid n, \\ 0, & p \nmid n. \end{cases} \tag{46.1}$$

and $Q_p = I - T_p$. Distinct T_p commute.

Proposition 46.1 (Prime-by-prime exact construction)**.** *On the finite interval $1 \le n \le X$,*

$$\mu_X = \prod_{p \le X} (I - T_p) e_1. \tag{46.2}$$

If $M_{\mathcal{P}}(x)$ denotes the cumulative sequence after a finite set $\mathcal{P}$ of prime operators has been applied, adjoining one new prime p gives

$$M_{\mathcal{P}\cup\{p\}}(x) = M_{\mathcal{P}}(x) - M_{\mathcal{P}}\left(\left\lfloor \frac{x}{p} \right\rfloor\right). \tag{46.3}$$

This makes precise the earlier idea of imposing arithmetic rigidity only through selected prime channels. For a predeclared integer r, the primes may be partitioned by their index modulo r:

$$\mathcal{P}_j^{(r)} = \{p_\ell : \ell \equiv j \pmod r\}, \qquad j = 0, \dots, r-1.$$

The r commuting channel operators together recover (46.2). Values such as $r = 2$ or $r = 5$ are legitimate experimental designs only if fixed before evaluating the arithmetic output; selecting r after inspection would reintroduce a tuning channel.

Remark 46.2 (Status of this route). This prime-channel construction does not replace the divisor-rank filtration of Section 44. It is retained as an exact side route for testing whether the Leslie representation reacts differently to sparse, predeclared subsets of arithmetic constraints.

47. Earlier combinatorial locality barriers retained in Version 9

An earlier line of the project studied Möbius through spaces of compatible ternary completions rather than through spectral geometry. Those results are retained because they clarify what kinds of arithmetic input can and cannot plausibly collapse the divisor tower. They are logically independent of the Leslie calculations above.

Let $\Omega_N(\mathcal{I})$ denote the family of length-N ternary assignments compatible with a declared information set $\mathcal{I}$, and let

$$\mathfrak{B}_N(\mathcal{I}) = \max_{\omega\in\Omega_N(\mathcal{I})} \left| \sum_{n\le N} \omega_n \right| \tag{47.1}$$

be the corresponding adversarial endpoint envelope. The reference space is the full set of 3^N words; every reduction below is measured relative to that reference.

Remark 47.1 (Distinct branches must not be merged). The previous work separated four genuinely different mechanisms:

(1) divisor-local constraints on independent variables;
(2) multiplicative prime-sign variables;
(3) residual closure after a finite prime set P_r;
(4) translational and global constraints.

Their coverage counts are not interchangeable. In particular, a statement proved for free prime signs cannot be transferred to independent divisor-local variables, and the residual P_r closure is a different model again.

Proposition 47.2 (Recorded locality barriers from the completion programme)**.** *Under the hypotheses used in the earlier completion calculations, the following lower-order coverage obstructions remain:*

(1) *divisor-local constraints on independent coordinates leave a linear-order family of upper squarefree degrees of freedom, of order* $\Omega(N)$*;*
(2) *free multiplicative prime signs retain a singleton-prime family of order* $\Omega(N/\log N)$*;*
(3) *bounded translational constraints of width* h *have elementary coverage only of order* $\Omega(N/h)$*;*
(4) *after closure under a finite prime set*

$$P_r = \{p \le p_r\},$$

the residual variables with least prime factor exceeding p_r *form a separate residual component, previously denoted through* P_r *or its residual size* $L_r(N)$*.*

These are coverage obstructions, not asymptotic statements about $M(N)$*.*

The same earlier analysis also found that imposing only squarefree support or only bare multiplicative sign structure does not by itself change the natural linear scale of the adversarial endpoint problem. In the notation used there,

$$\mathfrak{B}_N^{\mathrm{sqf}} = \mathfrak{B}_N^{\mathrm{mult}} = Q(N), \tag{47.2}$$

while fixing the sign at the prime 2 gives the exact reduction

$$\mathfrak{B}_N^{\mu(2)=-1} = Q_{\mathrm{odd}}(N) - Q_{\mathrm{odd}}(N/2). \tag{47.3}$$

The point of these formulas is structural: local arithmetic information can change constants without forcing the cancellation exponent sought in the global Mertens problem.

Remark 47.3 (Why the current filtration is not ruled out by these barriers). The elementary coverage arguments above apply to local or bounded-width information. They do not rule out aggregate constraints, full incidence relations, or hybrid arithmetic–translational observables that touch a macroscopically large part of the state at once. The present divisor filtration therefore remains compatible with those negative locality results: its exact extension block $B_k(X)$ is itself global across the new layer. What remains open is whether a *predeclared* observable of that global layer can satisfy the repaired Leslie normal constraints and then collapse to a low-dimensional continuation channel.

48. Earlier arithmetic constructions absorbed into the filtration

Several earlier experiments can now be placed inside the same divisibility framework. They are retained because they provide canonical state coordinates or controls, but none overrides the active observable-compression problem.

48.1. **Alternating divisor-chain expansion.** For fixed X, write

$$D_X = I + N_X,$$

where N_X is strictly lower triangular. Since N_X is nilpotent,

$$\boxed{D_X^{-1} = I - N_X + N_X^2 - N_X^3 + \cdots} \tag{48.1}$$

is a finite exact sum. The entry $(N_X^j)_{n,d}$ counts strict divisibility chains

$$d = d_0 \mid d_1 \mid \cdots \mid d_j = n, \qquad d_0 < d_1 < \cdots < d_j.$$

Consequently,

$$\mu(n/d) = \sum_{j\geq 0} (-1)^j \#\{d = d_0 < \cdots < d_j = n : \ d_i \mid d_{i+1}\}, \tag{48.2}$$

for divisor pairs $d \mid n$.

This gives a canonical graded arithmetic state without choosing an encoder. Define

$$c_j(n) = (N_X^j e_1)_n. \tag{48.3}$$

Then

$$\boxed{\mu(n) = \sum_{j\geq 0} (-1)^j c_j(n).} \tag{48.4}$$

Thus the chain-count state has a forced alternating observation vector $h_{\text{chain}} = (1, -1, 1, -1, \ldots)$. A natural Leslie question is whether the evolution of these graded chain counts, rather than the raw vector μ, admits a positive companion realization or a coherent low-dimensional observable.

48.2. **Nested completion spaces as the dual filtration.** A separate earlier programme added prime, prime-square, and divisibility constraints sequentially to the set of Möbius-compatible assignments. If $\mathfrak{C}_N^{(j)}$ denotes the assignments still compatible after stage j, then

$$\mathfrak{C}_N^{(0)} \supseteq \mathfrak{C}_N^{(1)} \supseteq \cdots .$$

The associated extremal quantity was

$$\mathcal{B}_N^{(j)} = \max_{\omega \in \mathfrak{C}_N^{(j)}} \left| \sum_{n\leq N} \omega_n \right|, \qquad \Delta_j(N) = \mathcal{B}_N^{(j-1)} - \mathcal{B}_N^{(j)}.$$

This is dual to the present ascending state filtration: one construction enlarges the exact arithmetic state while preserving old coordinates; the other shrinks the compatible solution space while preserving old constraints. The duality is conceptual only; no bound from the completion experiment is imported into the present Leslie argument.

48.3. **Weighted factorization-path state.** Another earlier construction retained the multiplicity of factorization paths instead of collapsing immediately to $\mu(n)$. With $e(x) = e^{2\pi i x}$, an edge $C \to Cp$ carried the lift

$$[C : p] = p\, e(C/p). \tag{48.5}$$

For a path $w = (p_{i_1}, \dots, p_{i_r})$ with partial products $C_j = C_{j-1}p_{i_j}$, its holonomy was

$$H(w) = e\left(\sum_{j=1}^{r} \frac{C_{j-1}}{p_{i_j}}\right), \qquad \widehat{n}(w) = nH(w). \tag{48.6}$$

The path spectrum, mean phase, and interference energy were

$$\mathcal{S}(n) = \{\widehat{n}(w)\}, \qquad \Psi(n) = \frac{1}{N(n)} \sum_{w} H(w), \qquad E(n) = 1 - |\Psi(n)|^2. \tag{48.7}$$

These quantities are not asserted to generate Möbius and are not used in any theorem of the current route. Their relevance is narrower: they are examples of arithmetic observables built from factorization paths before the Mertens output is inspected. They may therefore serve as predeclared candidates or controls when testing the observable $\mathcal{O}_{k,X}$, without changing the exact incidence-layer source.

49. Independent divisibility-surface coordinates imported into Version 9

A separate companion project developed a second finite coordinate system built from the same divisibility arithmetic. It is integrated here because it supplies canonical, parameter-free observables that may be relevant to the open map $\mathcal{O}_{k,X}$. None of the results in this section is used as evidence that Leslie compression exists.

49.1. **Threshold surface and exact first-difference inverse.** For $n > 1$ and $r \geq 1$, define the divisor-threshold surface

$$\mathcal{F}^{\mathrm{surf}}(n, r) = \sum_{\substack{a \mid n \\ a > r}} \mu(n/a). \tag{49.1}$$

Its vertical jump is exact:

$$\mathcal{F}^{\mathrm{surf}}(n, r-1) - \mathcal{F}^{\mathrm{surf}}(n, r) = \mathbf{1}_{r \mid n} \mu(n/r). \tag{49.2}$$

At the first threshold one recovers the Möbius column itself:

$$\boxed{\mathcal{F}^{\mathrm{surf}}(n, 1) = -\mu(n), \qquad n > 1.} \tag{49.3}$$

Let

$$\mathsf{B}_N^{\mathrm{surf}} = \left(\mathcal{F}^{\mathrm{surf}}(n, r)\right)_{2 \leq n \leq N,\ 1 \leq r \leq N-1}. \tag{49.4}$$

This matrix is integral, lower triangular, and unit diagonal. Hence

$$\det \mathsf{B}_N^{\mathrm{surf}} = 1. \tag{49.5}$$

Its inverse is particularly simple:

$$\boxed{\left((\mathsf{B}_N^{\mathrm{surf}})^{-1}\right)_{r,m} = \mathbf{1}_{m \mid r+1} - \mathbf{1}_{m \mid r}.} \tag{49.6}$$

Equation (49.6) is important for the present manuscript because it gives a canonical arithmetic compression that is neither fitted nor spectral: divisor incidence is first integrated into threshold coordinates and then exactly inverted by an ordinary first difference. Thus the surface supplies a natural family of candidates from which a Leslie-facing observable $\mathcal{O}_{k,X}$ could be constructed without inspecting future Mertens values.

The row norm is also arithmetically explicit. If

$$1 = d_1 < d_2 < \cdots < d_t = n, \qquad M_n(j) = \sum_{i=1}^{j} \mu(d_i),$$

then

$$\sum_{r=0}^{n} \left|\mathcal{F}^{\mathrm{surf}}(n,r)\right|^2 = \sum_{i=0}^{t-1} (d_{i+1} - d_i) M_n(t-i)^2, \qquad d_0 := 0. \tag{49.7}$$

In particular,

$$\sum_{r} \left|\mathcal{F}^{\mathrm{surf}}(p^a,r)\right|^2 = \varphi(p^a), \qquad \sum_{r} \left|\mathcal{F}^{\mathrm{surf}}(pq,r)\right|^2 = (p-1)(q+1) \tag{49.8}$$

for primes $p < q$. These formulas give exact scale information for any future use of surface rows as Leslie observables.

49.2. **Additive threshold transform and necklace coordinates.** For $n > 1$, define

$$P_n(z) = \sum_{d|n} \mu(d) z^{n/d}, \qquad R_n(z) = \sum_{r=1}^{n-1} \mathcal{F}^{\mathrm{surf}}(n,r) z^r. \tag{49.9}$$

Then the threshold coordinate has the exact polynomial encoding

$$\boxed{R_n(z) = \frac{P_n(z)}{z-1}.} \tag{49.10}$$

Thus, up to the conventional factor $1/n$, the surface row is the coefficient vector of a divided necklace polynomial. In particular,

$$R_n(1) = \varphi(n). \tag{49.11}$$

If the rows are zero-extended to length $N-1$, their discrete Fourier transform is obtained by evaluating this same polynomial at roots of unity. Hence Fourier transformation in the threshold coordinate closes algebraically inside a divisor-determined polynomial family rather than introducing a fitted spectral model.

This supplies another predeclared candidate for $\mathcal{O}_{k,X}$: one may test selected threshold-Fourier coefficients or fixed linear combinations of them, provided the selection is made before the Mertens output is inspected. No cancellation theorem follows from the polynomial identity alone.

49.3. **Gram coordinates, GCD kernel, and the independent GOE route.** Define

$$\mathsf{G}_N^{\mathrm{surf}} = (\mathsf{B}_N^{\mathrm{surf}})^{\mathsf{T}} \mathsf{B}_N^{\mathrm{surf}}. \tag{49.12}$$

It is an integral positive-definite unimodular matrix. Writing

$$\mathsf{H}_N^{\mathrm{surf}} = (\mathsf{G}_N^{\mathrm{surf}})^{-1},$$

one has the exact kernel

$$\begin{aligned} (\mathsf{H}_N^{\mathrm{surf}})_{r,q} = {} & \tau(\gcd(r+1,q+1)) - \tau(\gcd(r+1,q)) \\ & - \tau(\gcd(r,q+1)) + \tau(\gcd(r,q)). \end{aligned} \tag{49.13}$$

Equivalently, if

$$A_N(x,y) = \tau(\gcd(x,y)) - 1$$

and Δ_N is the ordinary forward-difference matrix, then

$$\boxed{\mathsf{H}_N^{\mathrm{surf}} = \Delta_N A_N \Delta_N^{\mathsf{T}}.} \tag{49.14}$$

The column Gram of the inverse,

$$\mathsf{Q}_N^{\mathrm{surf}} = (\mathsf{B}_N^{\mathrm{surf}})^{-\mathsf{T}} (\mathsf{B}_N^{\mathrm{surf}})^{-1}, \tag{49.15}$$

is similar to $\mathsf{H}_N^{\mathrm{surf}}$ and therefore has the same spectrum. Its entries admit an exact CRT description. In particular, for coprime indices

$$(\mathsf{Q}_N^{\mathrm{surf}})_{m,n} \in \{0, -1, -2\}, \tag{49.16}$$

and in general each entry decomposes into an explicit arithmetic mean plus a bounded periodic zero-mean fluctuation in N. Its support satisfies

$$\mathrm{nnz}(\mathsf{Q}_N^{\mathrm{surf}}) = O(N \log^3 N), \tag{49.17}$$

so the support density tends to zero even though arithmetic hub rows remain.

A separate manuscript conjectures Sine_1/GOE bulk eigenvalue statistics for $\mathsf{G}_N^{\mathrm{surf}}$ and studies finite-N spacing and eigenvector diagnostics. That conjecture remains logically independent here. In particular, GOE-like eigenvalue statistics, even if ultimately proved, would not imply the existence of a Leslie observable or a rank-one arithmetic-to-Leslie channel. Conversely, the exact identities (49.13)–(49.17) are usable arithmetic structures whether or not the GOE conjecture is true.

49.4. **Compact Gram-edge operator.** The same surface project derives an exact large-N edge scaling. For $y, z \in (0, 1]$, define

$$g(y, z) = \sum_{d,e \geq 1} \frac{\mu(d)\mu(e)}{[d, e]} (1 - \max(dy, ez))_+, \tag{49.18}$$

and let $\mathcal{G}$ be the integral operator on $L^2(0, 1)$ with kernel g. Then

$$\left\| N^{-2} \mathsf{G}_N^{\mathrm{surf}} - \mathcal{G} \right\|_1 \longrightarrow 0. \tag{49.19}$$

Consequently, if $a_1 \geq a_2 \geq \cdots > 0$ are the eigenvalues of $\mathcal{G}$, then for each fixed j,

$$\frac{\lambda_j(\mathsf{G}_N^{\mathrm{surf}})}{N^2} \longrightarrow a_j. \tag{49.20}$$

This is an independent example of a dimension-growing arithmetic matrix family with a coherent nontrivial spectral limit. It does not solve the Leslie tower problem, but it shows that dimension growth and spectral coherence can coexist without forcing a fixed finite spectrum.

50. Exact p-adic shell coordinates as predeclared arithmetic observables

Another companion construction gives canonical local coordinates prime by prime. Fix a prime p and define on $\mathbb{Z}_p \setminus \{0, -1\}$

$$\eta_p(x) = (-1)^{v_p(x) + v_p(x+1)}. \tag{50.1}$$

Let $\mathcal{F}_j$ be the residue-class σ-algebra modulo p^j, and put

$$M_j = \mathbb{E}(\eta_p \mid \mathcal{F}_j), \qquad \Delta_j = M_j - M_{j-1}.$$

The Haar mean is

$$\boxed{\int_{\mathbb{Z}_p} \eta_p \, dm_p = \frac{p-3}{p+1}.} \tag{50.2}$$

Thus $p = 3$ is the unique centered prime in this particular valuation-parity coordinate system.

The shell energies are exact:

$$\|\Delta_1\|_2^2 = \frac{8(p-2)}{(p+1)^2}, \tag{50.3}$$

and, for $j \geq 2$,

$$\boxed{\|\Delta_j\|_2^2 = \frac{8(p-1)}{(p+1)^2} p^{1-j}.} \tag{50.4}$$

Moreover,

$$M_0^2 + \sum_{j \geq 1} \|\Delta_j\|_2^2 = 1. \tag{50.5}$$

For $p = 3$ this specializes to

$$\|\Delta_1\|_2^2 = \frac{1}{2}, \qquad \|\Delta_j\|_2^2 = 3^{1-j} \quad (j \geq 2).$$

Remark 50.1 (Relevance and limitation). These shell variables are determined before Mertens output is inspected, so they are legitimate predeclared arithmetic coordinates for testing candidate observables. Their orthogonality is an $L^2(\mathbb{Z}_p)$ statement, not a cancellation theorem for integer sums. In particular, the known shell energies cannot be inserted into the Leslie programme as if different prime gauges were independent. A successful use would require an explicit causal map from the shell state to recurrence coefficients and then verification of the normal and rank-one conditions of Section 55.2.

51. Squarefree–unimodular branch closure

The companion p-adic project also isolates a rigidity mechanism that is structurally compatible with the philosophy of preserving old constraints. Let

$$M = \begin{pmatrix} q & A \\ r & B \end{pmatrix} \in SL_2(\mathbb{Z}), \qquad L_1(m) = qm + A, \quad L_2(m) = rm + B.$$

Suppose an admissible branch satisfies

$$d_1^2 \mid L_1(m), \qquad d_2^2 \mid L_2(m).$$

Then the determinant-one relation forces

$$(d_1, d_2) = 1, \qquad (d_1, q) = 1, \qquad (d_2, r) = 1.$$

CRT gives a unique residue class

$$m = c + d_1^2 d_2^2 t,$$

and after dividing out the two squares the renormalized affine forms are

$$\widetilde{L}_1(t) = q d_2^2 t + a, \qquad \widetilde{L}_2(t) = r d_1^2 t + b, \tag{51.1}$$

with

$$\boxed{\det \begin{pmatrix} q d_2^2 & a \\ r d_1^2 & b \end{pmatrix} = 1.} \tag{51.2}$$

This gives an exact branchwise example in which arithmetic square extraction adds constraints without destroying the governing determinant invariant. It is not a Leslie theorem, but it is a useful model for what "arithmetic enters without breaking the old geometry" should mean.

The same branch decomposition also yields an exact finite Möbius–Liouville transfer. For a finite interval I and the two affine forms above, define

$$C_f(M; I) = \sum_{m \in I} f(L_1(m)) f(L_2(m)).$$

Then the square-divisor expansion and the CRT branches give

$$C_\mu(M;I)=\sum_{d_1,d_2}\mu(d_1)\mu(d_2)C_\lambda(M_{d_1,d_2};I_{d_1,d_2}), \tag{51.3}$$

where only admissible coprime square-extraction branches contribute and every branch matrix M_{d_1,d_2} remains in $SL_2(\mathbb{Z})$. This identity is retained as an independent arithmetic route, not as an input to the active Leslie programme. In particular, it does not supply the missing decorrelation needed for ordinary cancellation, and Version 9 does not import external analytic machinery through this side identity.

52. Intrinsic congruence-divisor operator

The same surface factorization supplies a canonical Hermitian arithmetic operator independent of the Leslie fit. Let

$$\mathsf{B}_N^{\mathrm{surf}}=\mathsf{K}_N\mathsf{L}_N,\qquad \mathsf{J}_N=\mathsf{L}_N^{-1},$$

and define

$$\mathsf{A}_N=\mathsf{B}_N^{\mathrm{surf}}\mathsf{J}_N(\mathsf{B}_N^{\mathrm{surf}})^{-1},\qquad \mathsf{D}_N=\frac{\mathsf{A}_N-\mathsf{A}_N^*}{2i}. \tag{52.1}$$

For $2\le n,m\le N$, put

$$C(n,m)=\sum_{\substack{a\mid n\\ a\ge 2\\ a\equiv 1 \pmod m}}\mu(n/a). \tag{52.2}$$

Then

$$\boxed{(\mathsf{D}_N)_{n,m}=\frac{i}{2}(C(n,m)-C(m,n)).} \tag{52.3}$$

The kernel depends only on the indices, so D_N is a leading principal submatrix of D_{N+1} and its eigenvalues interlace across N.

The available unconditional norm bounds pin its polynomial order:

$$\left(\frac{\sqrt{6}}{2\pi}+o(1)\right)\sqrt{N}\le\lambda_{\max}(\mathsf{D}_N)\le O_\varepsilon(N^{1/2+\varepsilon}). \tag{52.4}$$

No Riemann zero, von Mangoldt weight, or externally supplied spectral variable enters (52.1). The construction is therefore another admissible source of arithmetic state variables for the current programme. It must nevertheless pass the same anti-tautology rule as every other candidate: any map from its spectral data to Leslie coefficients must be fixed before the target Mertens output is examined.

53. Coherent Leslie towers

The arithmetic filtration suggests that the Leslie state should be allowed to grow rather than forcing every level into one fixed finite-dimensional model.

Definition 53.1 (Coherent Leslie tower)**.** A coherent Leslie tower is a sequence of positive Leslie companion matrices

$$L_k\in\mathbb{R}^{m_k\times m_k},\qquad m_1<m_2<\cdots,$$

together with surjective maps

$$\Pi_k:\mathbb{R}^{m_{k+1}}\to\mathbb{R}^{m_k}$$

such that

$$\boxed{\Pi_kL_{k+1}=L_k\Pi_k.} \tag{53.1}$$

In coordinates adapted to $\ker \Pi_k$, the intertwining relation has the block form

$$\widetilde{L}_{k+1} = \begin{pmatrix} L_k & 0 \\ C_k & G_k \end{pmatrix}, \tag{53.2}$$

which is structurally parallel to the arithmetic extension (44.2). The old quotient dynamics is preserved exactly, while a new state layer may receive information from the old one.

The companion representation gives an exact polynomial criterion.

Theorem 53.2 (Polynomial divisibility and companion quotients)**.** *Let P_k and P_{k+1} be monic polynomials and let $C(P_k)$, $C(P_{k+1})$ denote their companion operators. If*

$$P_{k+1}(z) = P_k(z) R_{k+1}(z) \tag{53.3}$$

for a monic polynomial R_{k+1}, then there exists a surjective linear map Π_k satisfying

$$\Pi_k C(P_{k+1}) = C(P_k) \Pi_k.$$

Conversely, for cyclic companion operators, a surjective intertwining map implies that the minimal polynomial of the quotient divides the parent polynomial; hence $P_k \mid P_{k+1}$.

Proof. View the companion operator of a monic polynomial P as multiplication by z on the cyclic module $\mathbb{R}[z]/(P)$. If $P_k \mid P_{k+1}$, reduction modulo P_k defines a surjective module homomorphism

$$\mathbb{R}[z]/(P_{k+1}) \longrightarrow \mathbb{R}[z]/(P_k)$$

that intertwines multiplication by z. Conversely, the minimal polynomial of a quotient operator divides that of the parent. A companion operator is cyclic, so its minimal polynomial equals its characteristic polynomial. □

Remark 53.3 (New interpretation of the auxiliary multiplier)*.* In Versions 2, 6, and 7 the multiplier

$$R_q(z)$$

was introduced mainly as a static positivity repair. Equation (53.3) supplies a different possible interpretation: the multiplier can represent the genuinely new state layer of a coherent arithmetic extension. Under this interpretation old characteristic factors are not refitted or discarded; they divide the next polynomial exactly.

Warning 53.4 (Positivity remains nontrivial)*.* Polynomial divisibility alone does not guarantee that every $C(P_k)$ is a nonnegative Leslie matrix. The tower problem therefore contains two simultaneous exact requirements:

$$P_k \mid P_{k+1} \qquad \text{and} \qquad P_k(z) = z^{m_k} - \sum_{\ell=1}^{m_k} a_{k,\ell} z^{m_k-\ell}, \quad a_{k,\ell} \geq 0.$$

The auxiliary-completion LP of the earlier sections remains relevant, but it can no longer be used independently at every level; the multipliers must also obey arithmetic coherence.

Theorem 53.5 (Coordinate truncation obstruction for strict Leslie companions)**.** *Let L_D be a standard order-D Leslie companion and let $P : \mathbb{R}^D \to \mathbb{R}^d$ be projection onto the first $d < D$ coordinates. If there exists an order-d Leslie companion L_d with*

$$PL_D = L_d P, \tag{53.4}$$

then

$$a_{d+1} = \cdots = a_D = 0$$

and therefore

$$\boxed{P_D(z) = z^{D-d} P_d(z).} \tag{53.5}$$

Thus a tower that preserves old levels by literal coordinate truncation can add only zero eigenvalues.

Proof. The upper-right block of L_D must vanish under (53.4). In a Leslie companion that block consists precisely of the tail coefficients in the first row. The factorization (53.5) follows immediately from the characteristic polynomial. □

Remark 53.6 (Abstract quotient versus coordinate Leslie form). Theorem 53.2 still permits nontrivial factors R_{k+1} through an abstract quotient map. However, after changing to a basis adapted to that quotient, the parent operator need not retain the standard Leslie companion form or entrywise positivity. Hence nontrivial coherent spectral extension and strict coordinate Leslie nesting are different requirements. Version 9 does not identify them.

54. Arithmetic–Leslie compatibility problem

The active problem can now be stated without asking Leslie geometry to invent arithmetic.

For the arithmetic filtration define the exact extension

$$A_k x = \begin{pmatrix} x \\ -B_k x \end{pmatrix}, \qquad A_k : \mathbb{R}^{|I_k|} \to \mathbb{R}^{|I_{k+1}|}. \tag{54.1}$$

Then

$$\mu_{k+1} = A_k \mu_k \qquad \text{and} \qquad \operatorname{Res}_k A_k = I,$$

where Res_k restricts to the old ideal.

Definition 54.1 (Arithmetic-driven Leslie realization)**.** An arithmetic-driven Leslie realization of the filtration consists of

(1) a coherent positive Leslie tower (L_k, Π_k);
(2) arithmetic-to-Leslie encoders $E_k : \mathbb{R}^{|I_k|} \to \mathbb{R}^{m_k}$ determined by a single predeclared rule from the divisor incidence data;
(3) Leslie embeddings $J_k : \mathbb{R}^{m_k} \to \mathbb{R}^{m_{k+1}}$ with $\Pi_k J_k = I$;
(4) compatibility

$$\boxed{E_{k+1} A_k = J_k E_k;} \tag{54.2}$$

(5) induced observation and source vectors satisfying

$$h_k^{\mathsf{T}} E_k = \mathbf{1}^{\mathsf{T}}, \qquad E_k e_1 = v_k. \tag{54.3}$$

If (54.2) and (54.3) hold, then

$$h_k^{\mathsf{T}} E_k \mu_k = \mathbf{1}^{\mathsf{T}} \mu_k, \tag{54.4}$$

so the arithmetic observable is preserved by construction rather than fitted afterward.

Warning 54.2 (The encoder is a stronger global bridge). If E_k is allowed to be chosen freely after seeing μ_k or M_k, the construction is tautological. The central unsolved requirement of Version 8 is therefore:

$$\begin{gathered} E_k,\ J_k,\ R_{k+1} \text{ must be determined from } D_k,\ B_k \\ \text{by one fixed rule, before inspecting the Mertens output.} \end{gathered} \tag{54.5}$$

Finding such a rule while preserving Leslie positivity would solve a strong global realization problem. Version 9 does not require this full encoder before testing the weaker local bridge: the immediate open problem is the causal observable $\mathcal{O}_{k,X}$ and the rank-one tangent criterion of Theorem 55.4.

This formulation explains why the Version 4 negative result is useful rather than terminal. The missing ingredient was not another fitted frequency, another Perron value, or another empirical innovation law. It was a source of arithmetic constraints that is external to the scalar Mertens history but exact and parameter-free.

55. A low-dimensional compression question for each new arithmetic layer

The full divisor layer (44.8) may have large dimension. Leslie geometry would become informative if its effect could be represented by much fewer coordinates without changing old levels.

Let $\mathcal{G}_k$ denote the tangent space of the admissible Leslie extension at level k and let $\Delta_k^{\mathrm{arith}} = -B_k\mu_k$ be the exact new arithmetic layer. Let

$$\iota_{k+1} : \mathbb{R}^{|S_{k+1}|} \to \mathbb{R}^{|I_{k+1}|}, \qquad \iota_{k+1}y = \begin{pmatrix} 0 \\ y \end{pmatrix},$$

be the canonical embedding of the new layer into the full arithmetic state. After a fixed, predeclared encoding into the Leslie state space, define the best r-dimensional defect

$$\rho_{k,r} = \frac{\operatorname{dist}\Big(E_{k+1}\iota_{k+1}\Delta_k^{\mathrm{arith}}, \mathcal{G}_{k,r}\Big)}{\|E_{k+1}\iota_{k+1}\Delta_k^{\mathrm{arith}}\|}, \tag{55.1}$$

where $\mathcal{G}_{k,r}$ is a predeclared r-dimensional admissible subspace or tangent model.

The earlier one-innovation programme corresponds conceptually to $r = 1$, but Version 8 no longer assumes that $r = 1$ must work. The falsifiable question is instead whether

$$r_k \ll |S_{k+1}(X)|$$

can remain small under coherent arithmetic extension. A failure at $r = 1$ would therefore not force the entire arithmetic-driven tower to fail; it would measure the information dimension actually required by the new divisor layer.

55.1. The raw arithmetic layer is not uniformly low dimensional.

The dimension question can be settled negatively before any Leslie observable is introduced.

Theorem 55.1 (Unbounded rank of the incidence extension block)**.** *For every fixed $k \geq 1$,*

$$\boxed{\operatorname{rank} B_k(X) \geq \#\{p \leq X/2^k : \ p \text{ an odd prime}\}.} \tag{55.2}$$

Hence $\operatorname{rank} B_k(X) \to \infty$ *as* $X \to \infty$.

Proof. For every odd prime $p \leq X/2^k$, set $n_p = 2^k p$. Then $\Omega(n_p) = k + 1$, so $n_p \in S_{k+1}(X)$. Restrict B_k to the rows n_p and the columns indexed by those same odd primes. For distinct odd primes p, p' one has $p' \mid 2^k p$ if and only if $p' = p$. The resulting submatrix is the identity. □

Corollary 55.2 (No fixed finite-dimensional subspace contains the raw truncated layers)**.** *Embed all vectors $\mu|_{S_{k+1}(X)}$ in the coordinate space of the infinite layer. Each time X passes a new squarefree n with $\Omega(n) = k + 1$, the layer vector changes by $(-1)^{k+1}e_n$. Therefore the span of the truncated layer vectors has dimension equal to the number of squarefree elements encountered. No fixed finite-dimensional linear subspace contains all raw layers.*

This rules out the interpretation of $-B_k\mu_k$ itself as a scalar innovation. If compression exists, it must be produced by an observable.

55.2. Observable compression and the exact Leslie obstruction.

Let

$$\mathcal{O}_{k,X} : \mathbb{R}^{|S_{k+1}(X)|} \longrightarrow \mathbb{R}^D \tag{55.3}$$

be a causal, predeclared arithmetic-to-recurrence observable, and define

$$\delta a_k = \mathcal{O}_{k,X}(-B_k\mu_k). \tag{55.4}$$

The word "causal" here means that $\mathcal{O}_{k,X}$ is determined by the current divisor incidence data and the past Leslie state, not by future Mertens values or by post-selection after seeing the new layer output.

At a regular repaired Leslie point let

$$T_{a_k}\mathcal{M}^{\mathrm{rep}}_{b,K,q} \subset \mathbb{R}^D$$

be the tangent space and let

$$\pi_N : \mathbb{R}^D \to N_{a_k}\mathcal{M}^{\mathrm{rep}}_{b,K,q}$$

be the quotient to the normal space. By Corollary 32.3, this normal space has dimension $K+1$ at regular points.

Proposition 55.3 (Normal compatibility precedes scalar compression)**.** *A necessary first-order condition for the arithmetic perturbation to remain in the repaired Leslie family is*

$$\boxed{\pi_N(\delta a_k) = 0.} \tag{55.5}$$

Thus the $K+1$ codimension from Version 7 measures the number of structural normal constraints on the observable image*; it does not measure the information dimension of B_k.*

Now let Γ_k be the regular one-dimensional common-equation continuation branch and let g_k span its tangent line.

Theorem 55.4 (Exact scalar-compression criterion)**.** *For the actual Möbius layer, exact first-order scalar Leslie compression is equivalent to*

$$\boxed{\delta a_k \in \mathrm{span}\{g_k\}.} \tag{55.6}$$

If the same single channel is required to compress every *admissible input to B_k, then it is necessary and sufficient that*

$$\boxed{\mathrm{Im}(\mathcal{O}_{k,X}B_k) \subseteq \mathrm{span}\{g_k\},} \tag{55.7}$$

equivalently

$$\mathrm{rank}(\mathcal{O}_{k,X}B_k) \le 1 \tag{55.8}$$

with image tangent to the continued branch.

Remark 55.5 (Intrinsic innovation coordinate). An orthogonal projection of δa_k onto g_k depends on the chosen coefficient metric. The branch observation from Theorem 33.4 gives an intrinsic alternative. Normalize g_k by $d\chi_k(g_k) = 1$. For a tangent perturbation define

$$\boxed{\xi_k = d\chi_k(\delta a_k).} \tag{55.9}$$

The first-order failure of scalar Leslie compression is naturally represented by the pair

$$\boxed{(\pi_N \delta a_k, \delta a_k - \xi_k g_k)\,,} \tag{55.10}$$

separating failure to remain on the repaired variety from failure to remain on the one-dimensional continuation branch.

Question 55.6 (Fundamental Version 9 bridge)**.** Does there exist a causal, divisor-determined observable $\mathcal{O}_{k,X}$ for which the actual arithmetic forcing satisfies the normal compatibility conditions and has effective rank one on the relevant Leslie continuation, uniformly through an unbounded family of levels? Equivalently, can a complete high-dimensional divisor layer become one scalar Leslie innovation *only after observation*, even though neither B_k nor the raw layer is low dimensional?

In the concise form used for the active programme:

$$\boxed{\begin{array}{c}\text{determine whether there exists a causal } \mathcal{O}_{k,X} \text{ such that}\\ \mathcal{O}_{k,X}(-B_k\mu_k) \text{ satisfies the } K+1 \text{ normal Leslie constraints}\\ \text{and has effective rank 1 on the continued Leslie branch.}\end{array}} \tag{55.11}$$

If such an observable exists, one complete arithmetic layer becomes one scalar Leslie innovation after observation. If it does not, the connection between the divisibility tower and the Version 7 continuation theory fails precisely at this point.

Remark 55.7 (Precise status). Nothing in incidence algebra forces such an observable to exist, and the unbounded-rank theorem shows that it cannot arise from the raw layer by identity embedding. Conversely, the present results do not rule out a highly compressive causal observable. This is the exact point at which the divisibility tower either connects to the Version 7 scalar-continuation geometry or fails.

56. Concrete $X = 20$ arithmetic audit and contrast with a rigid Leslie matrix

For $X = 20$, the exact divisibility matrix is:

```
n= 1 | 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
n= 2 | 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
n= 3 | 1 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
n= 4 | 1 1 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
n= 5 | 1 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
n= 6 | 1 1 1 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0
n= 7 | 1 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0
n= 8 | 1 1 0 1 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0
n= 9 | 1 0 1 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0
n=10 | 1 1 0 0 1 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0
n=11 | 1 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0
n=12 | 1 1 1 1 0 1 0 0 0 0 0 1 0 0 0 0 0 0 0 0
n=13 | 1 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0
n=14 | 1 1 0 0 0 0 1 0 0 0 0 0 0 1 0 0 0 0 0 0
n=15 | 1 0 1 0 1 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0
n=16 | 1 1 0 1 0 0 0 1 0 0 0 0 0 0 0 1 0 0 0 0
n=17 | 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0
n=18 | 1 1 1 0 0 1 0 0 1 0 0 0 0 0 0 0 0 1 0 0
n=19 | 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0
n=20 | 1 1 0 1 1 0 0 0 0 1 0 0 0 0 0 0 0 0 0 1
```

Its exact inverse is:

```
n= 1 |  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
n= 2 | -1  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
n= 3 | -1  0  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
n= 4 |  0 -1  0  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
n= 5 | -1  0  0  0  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
n= 6 |  1 -1 -1  0  0  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0
n= 7 | -1  0  0  0  0  0  1  0  0  0  0  0  0  0  0  0  0  0  0  0
n= 8 |  0  0  0 -1  0  0  0  1  0  0  0  0  0  0  0  0  0  0  0  0
n= 9 |  0  0 -1  0  0  0  0  0  1  0  0  0  0  0  0  0  0  0  0  0
n=10 |  1 -1  0  0 -1  0  0  0  0  1  0  0  0  0  0  0  0  0  0  0
n=11 | -1  0  0  0  0  0  0  0  0  0  1  0  0  0  0  0  0  0  0  0
n=12 |  0  1  0 -1  0 -1  0  0  0  0  0  1  0  0  0  0  0  0  0  0
n=13 | -1  0  0  0  0  0  0  0  0  0  0  0  1  0  0  0  0  0  0  0
n=14 |  1 -1  0  0  0  0 -1  0  0  0  0  0  0  1  0  0  0  0  0  0
n=15 |  1  0 -1  0 -1  0  0  0  0  0  0  0  0  0  1  0  0  0  0  0
n=16 |  0  0  0  0  0  0  0 -1  0  0  0  0  0  0  0  1  0  0  0  0
n=17 | -1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0  0  0
n=18 |  0  0  1  0  0 -1  0  0 -1  0  0  0  0  0  0  0  0  1  0  0
n=19 | -1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0
n=20 |  0  1  0 -1  0  0  0  0  0 -1  0  0  0  0  0  0  0  0  0  1
```

The first column is exactly

$$(1, -1, -1, 0, -1, 1, -1, 0, 0, 1, -1, 0, -1, 1, 1, 0, -1, 0, -1, 0)^{\mathsf{T}},$$

and

$$\mathbf{1}^{\mathsf{T}} D_{20}^{-1} e_1 = -3 = M(20).$$

For comparison, a rigid critical Leslie matrix with

$$K = 9, \qquad m = 19, \qquad b = 1.15, \qquad r = \sqrt{b} = 1.0723805295$$

has first-row coefficients

$$a_\ell = r^\ell (r-1), \quad 1 \le \ell \le 18, \qquad a_{19} = r^{20},$$

numerically

0.077619	0.083238	0.089262	0.095723	0.102652	0.110082
0.118050	0.126594	0.135757	0.145583	0.156120	0.167421
0.179539	0.192534	0.206469	0.221414	0.237440	0.254626
			4.045558		

with the usual unit subdiagonal. Its Perron eigenvalue is 1.15 and the other 18 eigenvalues have modulus $\sqrt{b}$ to numerical error about 1.78×10^{-15}.

Remark 56.1 (The structural contrast). The two matrices have comparable finite size but opposite epistemic roles. D_{20} has no fitted parameter: divisibility fixes every entry, and its inverse contains Möbius literally. The rigid Leslie matrix realizes its prescribed spectrum to machine precision but has no arithmetic content unless arithmetic is supplied externally. Version 4 showed that spectral precision alone does not repair this deficit. Version 8 therefore asks for an exact structural map from the first object into a coherent tower of the second.

57. Version 8 route map retained as historical status

The accumulated programme now separates three routes.

`[CLOSED]` *Arithmetic-free fixed-family prediction.* The frozen Version 4 protocol is negative: exact membership is absent and the corrected approximate dynamics does not outperform the relevant controls.

`[ADAPT]` *Divisor-rank filtration.* The classical stratification by $\Omega(n)$ is used only as an exact nested order-ideal decomposition. The new question is coherence and low-dimensional Leslie representation of the layer maps $-B_k\mu_k$. No literature novelty claim is made here.

`[SIDE]` *Predeclared prime channels.* Exact operators $I - T_p$ may be grouped by prime index classes such as one channel out of every two or five. This does not override the filtration route; it is a controlled way to probe how sparse arithmetic restrictions enter a Leslie representation.

The active target is not an asymptotic estimate at this stage. It is the existence or nonexistence of a parameter-free commuting bridge

$$(D_k, B_k) \longrightarrow (E_k, J_k, R_{k+1}, L_k)$$

that preserves old arithmetic constraints, retains Leslie positivity, and quantifies the dimension of genuinely new information at each level.

58. Decision criterion for the Leslie programme

Conjecture 58.1 (Data-compatible repaired trajectory)**.** *There exist fixed integers $K \ge 1$ and $q \ge 0$, together with a fixed or predeclared past-only Perron rule, such that a persistent family of Mertens windows admits data-compatible repaired Leslie recurrence polynomials in $\mathcal{M}^{\mathrm{rep},+}_{b,K,q}$ and the corresponding regular critical-stride continuations predict the remaining $K+1$ new values from one revealed scalar innovation better than predeclared null and baseline models without recalibration.*

Remark 58.2 (Version 8 status of Conjecture 58.1). The conjecture is retained unchanged as part of the Version 7 record. Under the frozen Version 4 protocol $(K, b) = (4, 1.02)$ and $q \le 6$, its tested arithmetic-free realization is negative: strict membership is absent and the approximate predictive layer fails the relevant out-of-sample comparisons. The conjecture is therefore no longer the active

route in Version 8. This status is protocol-specific and is not promoted to a universal impossibility theorem for all Leslie constructions.

Remark 58.3 (Falsification). For a fixed protocol, the conjecture fails if exact or tolerance-certified membership is sporadic, the auxiliary degree or Perron rule must be repeatedly retuned, branch continuation is not causal and unique, the repaired overlap fails to be effectively one-dimensional, the first-new-value map loses observability, or the out-of-sample forecast does not improve on the predeclared controls.

Remark 58.4 (What a positive result would and would not mean). Persistent success would establish a strong dynamic redundancy property of the chosen Mertens observable: conditioned on the past state, one newly revealed scalar would determine the next $K+1$ values. It would not by itself produce a zero-latency law for Mertens. That stronger goal requires an additional arithmetic law or restriction for the innovation sequence (ξ_n).

Remark 58.5 (Pre-Möbius validation gate). Before Conjecture 58.1 is tested again, the repaired membership, continuation, branch tracking, and forecast routines should be validated on exact synthetic examples, including cases with known auxiliary degree and deliberately introduced rank, positivity, and observability failures. Numerical outputs should distinguish `PASS`, `FAIL`, `NOT_APPLICABLE`, and `NUMERICALLY_AMBIGUOUS`.

59. Conclusions

Proposition 59.1 (Version 7 structural conclusion)**.** *The exact moving-window theory now has four distinct structural layers.*

1. *Rigid critical Leslie spectra possess the coefficient, strain, positivity, and modal geometry developed in the earlier sections.*
2. *For fixed b, rigid coefficient vectors obey the exact reciprocal–Perron affine constraints, which yield the affine critical-stride scalarization at $s_* = K+2$.*
3. *Static positive completion of a prescribed rigid core is exactly the Toeplitz feasibility problem* (29.8)*, but this completion is not automatically a recurrence of the data.*
4. *For fixed (K, q, b), the data-compatible repaired coefficient family has regular codimension $K+1$ independently of q. Exact overlap of order-D data windows therefore has the same generic critical stride $s_* = K+2$; at that stride the repaired continuation is locally a one-dimensional curve. Under first-sample transversality, one newly observed scalar selects the local next coefficient vector and the remaining $K+1$ new values are causally determined.*

These are geometric and algebraic statements. They do not assert that Mertens windows satisfy the repaired membership conditions persistently.

Remark 59.2 (Corrected status of the numerical evidence). The rigid $K=7$ experiment remains a valid negative statement only for its eight evaluable frequency-estimator configurations. The 5 September 2026 v1.1 Mertens audit gives a separate negative baseline for the approximate projected protocol: zero strict rigid coverage at the selected (K, b) and much worse causal forecast error than AR. Its auxiliary $q \le 4$ obstruction applies to projected cores, not to exact data-compatible repaired recurrences. The exact symbolic gcd calculation on two raw Mertens windows is a cleaner rigid obstruction for those two windows. None of these results establishes or refutes the repaired Conjecture 58.1; that conjecture requires the corrected recurrence-first protocol of Section 38.

Remark 59.3 (Remaining arithmetic problem). If the repaired one-sample forecast succeeds persistently, Leslie geometry will have reduced each critical block to one external scalar innovation but will not have supplied a law for the innovation sequence. A zero-latency characterization therefore requires genuinely arithmetic information about (ξ_n) or an equivalent state forcing. Conversely, failure of the one-sample forecast already rules out the proposed dynamic compression without needing any theory of ξ_n.

60. Version 8 synthesis retained for comparison

Proposition 60.1 (What is now exact)**.** *The following statements require no empirical fit:*

1. *finite divisibility inversion $D_X^{-1}(n,d) = \mu(n/d)$ on divisor pairs;*
2. *the forced arithmetic realization $D_X \mu_X = e_1$ with $h = \mathbf{1}$ and $v = e_1$;*
3. *the divisor-rank block extension $D_{k+1} = \left(\begin{smallmatrix} D_k & 0 \\ B_k & I \end{smallmatrix}\right)$;*
4. *coherent arithmetic restriction $\mu_{k+1}|_{I_k} = \mu_k$ and new layer $-B_k\mu_k$;*
5. *prime-by-prime finite construction by commuting operators $I - T_p$;*
6. *the companion-module fact that $P_k \mid P_{k+1}$ is equivalent to the existence of the corresponding cyclic quotient intertwining.*

None of these statements supplies a positive Leslie tower by itself.

Conjecture 60.2 (Arithmetic-driven coherent Leslie tower)**.** *There exists a fixed, parameter-free rule depending only on the finite divisor incidence data (D_k, B_k) that produces encoders, embeddings, and positive Leslie characteristic multipliers so that the commuting relation*

$$E_{k+1}A_k = J_k E_k$$

and the induced observable constraints

$$h_k^{\mathsf{T}} E_k = \mathbf{1}^{\mathsf{T}}, \qquad E_k e_1 = v_k$$

hold coherently through an unbounded sequence of levels.

Question 60.3 (Information dimension of a divisor layer)**.** If Conjecture 60.2 is too strong, what is the smallest predeclared dimension r_k required to represent the exact layer $-B_k\mu_k$ inside the admissible Leslie extension with a controlled defect? Does r_k remain bounded, grow slowly, or scale with the layer size?

Remark 60.4 (What would constitute genuine progress)*.* A useful result is no longer a low regression error obtained after fitting Mertens. It is one of:

1. an explicit divisor-determined construction of the multiplier R_{k+1} and the intertwiner Π_k preserving positivity;
2. a proof that no such positive coherent extension exists beyond a stated class;
3. a nontrivial bound on the minimal information dimension r_k of the new arithmetic layer;
4. an exact structural invariant of the incidence blocks B_k that transports through the Leslie tower.

Each target is falsifiable without introducing zeta functions or asymptotic smoothing.

61. Version 9 synthesis: exact obstructions and the surviving bridge

Version 9 separates the source, the observable, and the Leslie geometry.

Proposition 61.1 (What the new arithmetic analysis rules out)**.** *For fixed $k \geq 1$, the raw incidence block $B_k(X)$ has unbounded rank as $X \to \infty$, and the truncated Möbius layers span spaces of unbounded dimension. Therefore no interpretation in which the complete new divisor layer is itself a fixed-dimensional, and in particular scalar, Leslie innovation can be correct.*

Proposition 61.2 (What survives from the Version 7 geometry)**.** *At a regular repaired point the coefficient family has an intrinsic normal space of dimension $K+1$. A divisor-driven perturbation must first annihilate these $K+1$ normal obstructions. If it does so and the overlap fiber is a regular curve, exact one-scalar continuation requires its remaining tangent image to lie in one line. Thus Version 7 survives as a geometric test on the* observable effect *of arithmetic, not as a statement about the dimension of arithmetic itself.*

Proposition 61.3 (What a strict nested Leslie tower cannot do)**.** *If old Leslie levels are preserved by literal projection onto the first coordinates of a larger standard companion, the only new characteristic factors are powers of z. Nonzero auxiliary modes therefore require an abstract quotient/intertwining construction or a different state realization; coordinate nesting alone cannot implement the desired arithmetic growth.*

Question 61.4 (Central open problem)**.**

Determine whether there exists a causal divisor-determined observable $\mathcal{O}_{k,X}$ such that $\mathcal{O}_{k,X}(-B_k\mu_k)$ satisfies the $K+1$ normal Leslie constraints and is confined to the one-dimensional continuation tangent, equivalently giving effective rank one on the relevant accessible arithmetic state.

If such an observable exists, one complete arithmetic layer becomes one Leslie innovation only after observation. If it does not, the connection between the divisibility filtration and the Version 7 scalar-continuation theory fails exactly at this interface.

Remark 61.5 (Route hierarchy)*.* The active hierarchy is now

$$\text{divisor incidence layer} \longrightarrow \text{canonical or predeclared observable}$$
$$\longrightarrow \text{normal Leslie compatibility} \longrightarrow \text{one-dimensional continuation test.}$$

The divisor-chain state (48.3), the nested completion filtration, the weighted factorization-path observables, the predeclared prime channels, the threshold-surface coordinates (49.1)–(49.6), the exact p-adic shell state (50.4), and the intrinsic congruence-divisor operator (52.3) are retained as arithmetic-side constructions or controls. None is allowed to replace the rank and transversality conditions above by post-hoc fitting.

Notebook History

Version 1.0: static rigid Leslie family and spectral comparison.
Version 2.0: positive spectral completion with auxiliary eigenvalues.
Version 3.0: moving rigid Leslie matrices on consecutive windows, corrected tangent-cone formulation, first- and second-order spectral transport, and a falsifiable forecasting protocol.
Version 4.0: exact strain coordinates, minimum Perron value, universal critical region, phase Jacobian, viability conditions, and modal amplitude transport.
Version 5.0: fixed-Perron reciprocal affine hull, exact overlap rigidity, critical stride $K+2$, scalar innovation transport, admissible innovation intervals, direct determinant tests, and the closed $K=1$ model.
Version 6.0: negative rigid $K=7$ margin test, exact minimal auxiliary-completion index, Toeplitz LP formulation, Farkas certificates of lower-degree impossibility, and the monotone auxiliary defect profile.
Version 7.0: recurrence-first data compatibility, separation of static completion from dynamic realization, exact local repaired mobility with codimension independent of auxiliary degree, nonlinear one-scalar continuation at $s_* = K+2$, explicit hidden-information accounting, correction of the previous approximate scalarization, and a frozen reinterpretation of the 5 September 2026 numerical audit.
Version 8.0: closure of the frozen arithmetic-free predictive route after the corrected Version 4 audit; lifted-state treatment of periodicity; exact finite divisibility realization with forced $h = \mathbf{1}$ and $v = e_1$; coherent Ω-filtration and block extension $-B_k\mu_k$; exact prime-injection side route; reinterpretation of auxiliary polynomial factors as possible new arithmetic layers; and formulation of the arithmetic-driven coherent Leslie tower problem.
Version 8.1: the finite-range quantity $\max|M(x)|/\sqrt{x}$ is explicitly demoted from "baseline" to historical diagnostic only; known limsup/liminf results are recorded to prevent any finite-range envelope interpretation, and the diagnostic is assigned no evidential weight in the arithmetic-driven

route.

Version 9.0: exact kernel and critical resultant of the repaired multiplication map; intrinsic normal-space codimension and augmented-rank observability; global branch counting, irreducibility and finite-window minimality; canonical divisor-layer continuation; unbounded-rank obstruction for $B_k(X)$ and for raw Möbius layers; coordinate-projection obstruction for strict Leslie towers; recovery of divisor-chain, nested-completion, weighted factorization-path, and earlier locality-barrier constructions; integration of the independent threshold-surface/unimodular, Gram/CRT, compact-edge, p-adic-shell, squarefree-SL_2, and intrinsic congruence-divisor operator routes; and reformulation of the active bridge as a causal observable $\mathcal{O}_{k,X}$ whose image must satisfy the $K+1$ normal Leslie constraints and collapse to the one-dimensional continuation tangent.

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